

Characterizing E.M.A. Compliance.

Using machine learning methods to investigate relationships related to participant compliance on Ecological Momentary Assessments (EMA).

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0 Definitions

For this semester's analysis, we measured participant compliance in three different ways: *Standard Compliance*, *Attempt Compliance*, and *Full Compliance*.

- *Standard Compliance* is defined as the total number of questions answered divided by the total number of questions asked during a time period.
- *Attempt Compliance* is defined as the total number of surveys attempted divided by the total number of surveys given during a time period. A survey was considered "attempted" if at least one question was answered.
- *Full Compliance* is defined as the total number of surveys fully completed divided by the total number of surveys given during a time period. A survey was considered "fully completed" if all questions (including open prompts) were answered.

All three forms of compliance were calculated at a daily, weekly, and study period level (replace "*time period*" in the definitions above with either "*day*", "*week*", or "*study*"). We used the level of compliance most appropriate for each feature.

Abstract (Summary)

1 *Stable Features & Compliance.*

Across Life@CMU Phase 1 and Phase 2, there was no significant relationship between EMA compliance and pre-semester psychological variables. The variables we tested were:

- Big Five Inventory (BFI).
- Interpersonal Support Evaluation List (ISEL).
- Sensation Seeking Scale (SSS).
- Center for Epidemiologic Studies - Depression Scale (CESD).
- Children's Interview for Psychiatric Syndromes (CHIPS).
- Brief Young Adult Alcohol Consequences Questionnaire (BYAACQ).
- Perceived Stress Scale (PSS).
- Beck Depression Inventory (BDI).
- Mindful Attention Awareness Scale (MAAS).

The only significant relationship we found was between the *compliance outliers* (compliance > 80% or < 20%) and some Big Five Personality traits. We describe the relationships and *p*-values below from most to least significant:

- BFI-Conscientiousness (**statistically significant**).
 - *direction*: positive (higher conscientiousness, higher compliance).
 - *p*-value: **0.0004**
- BFI-Extraversion (**statistically significant**).
 - *direction*: negative (higher extraversion, lower compliance).
 - *p*-value: **0.017**
- BFI-Openness.
 - *direction*: negative (higher openness, lower compliance).
 - *p*-value: 0.304.
- BFI-Neuroticism.
 - *direction*: positive (higher neuroticism, higher compliance).
 - *p*-value: 0.440.
- BFI-Agreeableness.
 - *direction*: positive (higher agreeableness, higher compliance).
 - *p*-value: 0.456.

If one fits machine learning models to predict compliance from the above psychological factors, the only features the models rely on the most are Big Five Personality.

2 Momentary Features & Compliance.

2.1 Fitbit Data & Compliance.

With Life@CMU Phase 1 Fitbit sleep data, we investigated the relationship between sleep patterns and compliance. The variables we tested are:

- *Time individuals go to bed (bedtime)*
- *Time individuals go to bed on weekends (bedtime_weekend)*
- *Time individuals go to bed on weekdays (bedtime_weekday)*
- *Variation of time individuals go to bed (bedtime_std)*
- *Variation of successive time individuals go to bed (bedtime_mssd)*
- *Median of variation of successive time individuals go to bed (bedtime_mssd_median)*
- *Time individuals wake up (waketime)*
- *Time individuals wake up on weekends (waketime_weekend)*
- *Time individuals wake up on weekdays (waketime_weekday)*
- *Variation of successive time individuals wake up (waketime_mssd)*
- *Median of variation of successive time individuals wake up (waketime_mssd_median)*
- *Midpoint of time individuals are sleeping (midpoint_sleep)*
- *Midpoint of time individuals are sleeping on the weekends (midpoint_sleep_weekend)*
- *Midpoint of time individuals are sleeping on the weekdays (midpoint_sleep_weekday)*
- *Variation of successive midpoint of time individuals are sleeping (midpoint_sleep_mssd)*
- *Median of variation of successive midpoint of time individuals are sleeping (midpoint_sleep_mssd_median)*
- *Average length of naptime (avg_nap_length)*
- *Average time in bed (time_in_bed)*
- *Average time individuals are asleep (TST)*
- *Variation of time individuals are asleep (TST_std)*
- *Average time awake in bed (WASO_fraction)*
- *Average time restless in bed (restless_fraction)*
- *Average sleep time by 24 hours (avg_24_hr_sleep)*
- *Fraction of total nap time over total sleep time (frac_napped_of_total_sleep)*
- *Fraction of number of naps over number of sleep (frac_sleep_episodes_as_naps)*
- *Number of naps (num_naps)*
- *Social jetlag [average time individuals go to bed on weekend - average time individuals go to bed on weekday] (social_jetlag)*

We found strong significant negative relationships between all types of compliance (Standard, Attempt, Full) and:

- *Time individuals go to bed (bedtime)*

- Standard Compliance
 - *p-value*: **0.000102**
 - *correlation coefficient*: -0.35889, *direction*: negative
- Attempt Compliance
 - *p-value*: **0.0000979**
 - *correlation coefficient*: -0.35975, *direction*: negative
- Full Compliance
 - *p-value*: **0.00738**
 - *correlation coefficient*: -0.25190, *direction*: negative
- Time individuals wake up (*waketime*)
 - Standard Compliance
 - *p-value*: **0.000498**
 - *correlation coefficient*: -0.32371, *direction*: negative
 - Attempt Compliance
 - *p-value*: **0.000518**
 - *correlation coefficient*: -0.32280, *direction*: negative
 - Full Compliance
 - *p-value*: **0.00164**
 - *correlation coefficient*: -0.00081, *direction*: negative
- Midpoint of time individuals are sleeping (*midpoint_sleep*)
 - Standard Compliance
 - *p-value*: **0.0000445**
 - *correlation coefficient*: -0.37577, *direction*: negative
 - Attempt Compliance
 - *p-value*: **0.0000445**
 - *correlation coefficient*: -0.37578, *direction*: negative
 - Full Compliance
 - *p-value*: **0.00135**
 - *correlation coefficient*: -0.29923, *direction*: negative

...where a *strong relationship* refers to a statistically significant correlation with all three types of compliance (having outliers removed). We removed the feature outliers of individuals who deviate 3 standard deviations away from the mean after conducting exploratory data analysis.

We found possible negative relationships between compliance and:

- Variation of time individuals go to bed (*bedtime_std*)
 - Full Compliance
 - *p-value*: **0.0274**
 - *correlation coefficient*: -0.20850, *direction*: negative
- Variation of successive time individuals wake up (*waketime_mssd*)

- Full Compliance
 - *p-value*: **0.0133**
 - *correlation coefficient*: -0.23339, *direction*: negative
- Average length of naptime (*avg_nap_length*)
 - Attempt Compliance
 - *p-value*: **0.0206**
 - *correlation coefficient*: -0.26352, *direction*: negative
 - Standard Compliance
 - *p-value*: **0.0202**
 - *correlation coefficient*: -0.26420, *direction*: negative

...where a *possible relationship* refers to a statistically significant correlation between at least one type of compliance and a variable (having outliers removed).

We found no significant relationship between all types of compliance and:

- Average time in bed (*time_in_bed*)
- Average time individuals are asleep [*average time in bed* - *average time awake in bed* - *average time restless in bed*] (*TST*)
- Variation of time individuals are asleep (*TST_std*)
- Average time awake in bed [*average of (time awake in bed / time in bed)*] (*WASO_fraction*)
- Average time restless in bed [*average of (time restless in bed / time in bed)*] (*restless_fraction*)
- Average sleep time by 24 hours [*total sleep time / total days*] (*avg_24_hr_sleep*)
- Fraction of nap time over sleep time [*total nap time / total sleep time*] (*frac_napped_of_total_sleep*)
- Fraction of number of naps over number of sleep [*total number of naps / total number of sleep*] (*frac_sleep_episodes_as_naps*)
- Variation of successive time individuals go to bed (*bedtime_mssd*)
- Variation of successive midpoint of time individuals are sleeping (*midpoint_sleep_mssd*)
- Median of variation of successive time individuals wake up (*waketime_mssd_median*)
- Median of variation of successive time individuals go to bed (*bedtime_mssd_median*)
- Median of variation of successive midpoint of time individuals are sleeping (*midpoint_sleep_mssd_median*)
- Number of naps (*num_naps*)
- Social jetlag [*average time individuals go to bed on weekend* - *average time individuals go to bed on weekday*] (*social_jetlag*)

...having outliers removed.

This result suggests that Individuals who go to bed later, who wake up later, and have disruptions in sleep patterns have lower compliance. And considering the fact that time individuals are asleep and average sleep time in 24 hours has no relationship with compliance, time when individuals go to bed and time when individuals wake up have stronger relationships with compliance.

2.2 *Smartphone Data & Compliance.*

We found no significant relationships between any type of compliance (Standard, Attempt, Full) and Life@CMU Phase 1 smartphone features. The smartphone features we tested with are *mean_screen*, *mean_unlocks*, and *mean_minutes*. *mean_screen* is the average number of screen-on events over the week (averaging the hourly counts), *mean_unlocks* is the average number of unlock events over the week (averaging the hourly counts), and *mean_minutes* is average number of time spent on one's phone over the week (averaging the hourly means).

This result suggests that smartphones and compliance have no relationship and may be a good medium for monitoring EMA surveys, as we found that smartphones and compliance are not correlated.

3 *Contextual Features & Compliance.*

3.1 *Survey Quality & Compliance*

We calculated **Scale Point Variation** as a measure of straightlining survey behavior with Life@CMU Phase 1 responses. We calculated scale point variation with respect to *Affect* questions, as it meets the criteria of a battery of questions that are in the same question category, have the same scale points, asked four times a day, have the same response variables, and consists of more than two questions.

We found no significant relationship between *Affect* straightlining questions and any type of compliance. However, we found a statistically significant negative correlation with:

- Standard deviation of *Affect* straightlining and all types of compliance
 - Standard Compliance
 - **p-value: 0.0000488**
 - *correlation coefficient: -0.30622, direction: negative*
 - Attempt Compliance
 - **p-value: 0.0000459**
 - *correlation coefficient: -0.30725, direction: negative*
 - Full Compliance

- ***p-value: 0.00236***
- *correlation coefficient: -0.23178, direction: negative*

This result suggests that individuals with higher variation of straightlining over the course of the study have lower compliance. In other words, individuals with varying levels of straightlining (sometimes straightline and sometimes not) have lower compliance.

3.2 *Survey Time & Compliance*

If we remove skipped surveys, there is no significant relationship between the median length of time a participant spends on a survey and their compliance (*p*-value: 0.394, *coef*: -0.0001) .

If we let skipped surveys have time 0, there is a strong inverse quadratic relationship (upside-down U: extremely fast and slow survey takers have low compliance; individuals with average survey times have high compliance) between the median time a participant spends on a survey and their compliance (*p*-value: 7.87×10^{-7}).

1 Stable Features & Compliance.

Our group has done substantial work in this space over the past year. You can find these results in this document:

<https://docs.google.com/document/d/1V0UmEV6XRCq4jaYM-PoanAFdQiRAVsLRV6dR1fySO5Y/edit?usp=sharing>

2 Momentary Features & Compliance.

2.1 *Fitbit Data and Compliance.*

Abstract

Sleep is an essential part of our lives. With sleep patterns, we can understand individuals' daily activities, including their compliance. We investigated the relationship between sleep patterns and compliance. We found significant relationships between characteristics of sleep and compliance rates with sleep data collected from Fitbit during Life@CMU Phase 1.

We found *strong relationships* between all types of compliance (Standard, Attempt, and Full) - and time individuals go to bed, time individuals wake up, and midpoint of time individuals are sleeping. A *strong relationship* refers to a statistically significant correlation with all three types of compliance (having outliers removed). We found *possible relationships* between compliance and variation of time individuals go to bed, variation of time individuals wake up, and length of naptime with compliance. A *possible relationship* refers to a statistically significant correlation between at least one type of compliance and a variable. The overall results suggest that understanding sleep patterns could predict compliance of an individual.

Introduction

Fitbit is a wireless-enabled wearable device that monitors heart rate, quality of sleep, and steps taken. The Life@CMU Phase 1 study required participants to wear a Fitbit, which collected individuals' steps and sleep data. Using this passively collected Fitbit data, we aimed to understand each participant's daily routine pattern of sleep and its relationship with different definitions of compliance.

Past study that explores factors influencing compliance with data collected with Fitbit exists. But no previous literature has looked at the relationship between sleep and compliance with sleep data collected with Fitbit. In our analysis, we found statistically significant relationships between sleep features and different definitions of compliance.

Method

Sleep data collected with Fitbit consists of 28 features excluding participant ID. The features include measures of number of naps, average nap length, number of sleep, average sleep hours, bedtime, waketime, bedtime on weekdays, bedtime on weekends, waketime on weekdays, waketime on weekends, midpoint sleep time on weekdays and weekends, and other sleep features that measure standard sleep routines of a participant. We computed the mean of each sleep feature over three weeks for each individual participant. And we also computed the mean of different definitions of compliance (Full, Attempt, and Standard) over all three weeks. We joined mean sleep features with mean compliance on participant ID, and we ran simple linear regression of all sleep features against each definition of compliance. As we ran simple linear regression models, we detected outliers that were affecting the statistical significance relationship between compliance and each feature, as shown in **Figure 2.1.1** and **Figure 2.1.2** for *bedtime_mssd* and *waketime_mssd*. **Figure 2.1.3** and **Figure 2.1.4** show distributions of *bedtime_mssd* and *waketime_mssd* that have clear outliers.

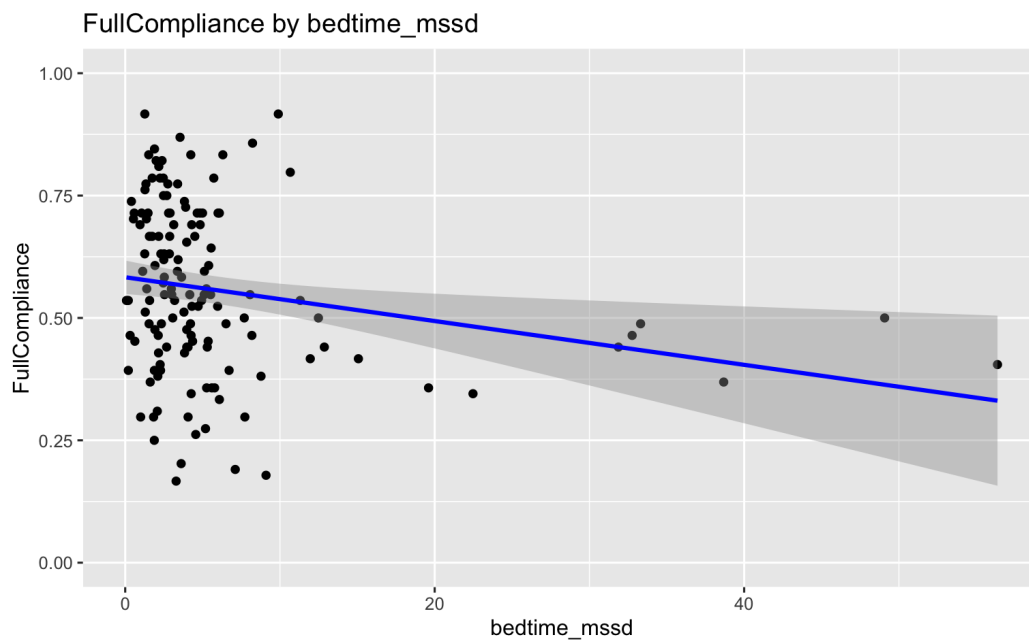


Figure 2.1.1

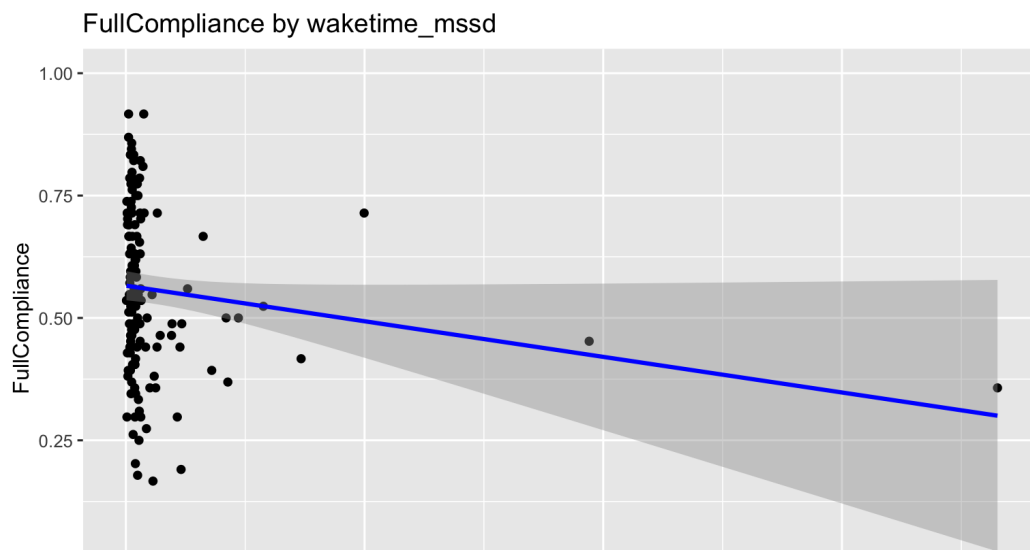
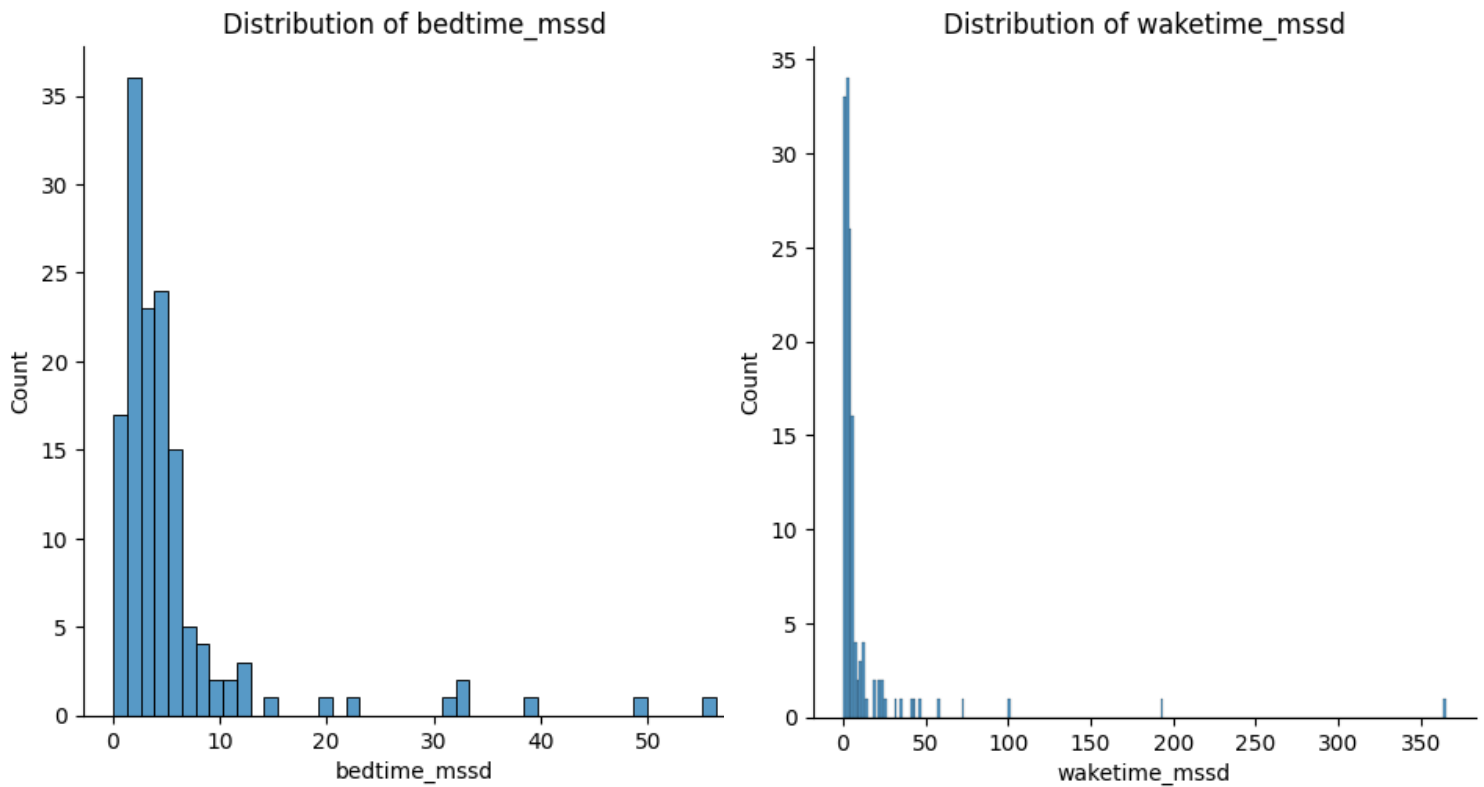


Figure 2.1.2



Left: **Figure 2.1.3**

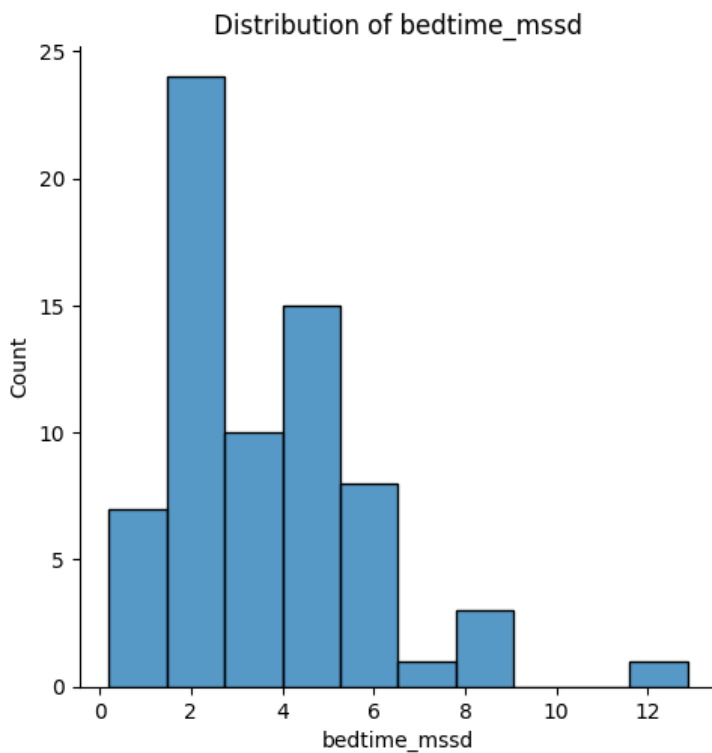
Right: **Figure 2.1.4**

Hence, we removed outliers that deviate 3 standard deviations from the mean for features that were affecting the statistical significance relationship between compliance and had clear outliers based on the regression plots and univariate distribution plots.

The features we removed outliers with are *bedtime_mssd*, *bedtime_mssd_median*, *bedtime_std*, *bedtime_weekend*, *waketime_mssd*, *waketime_mssd_median*, *waketime_weekend*, *midpoint_sleep_mssd*, *midpoint_sleep_mssd_median*, and *midpoint_sleep_weekend*.

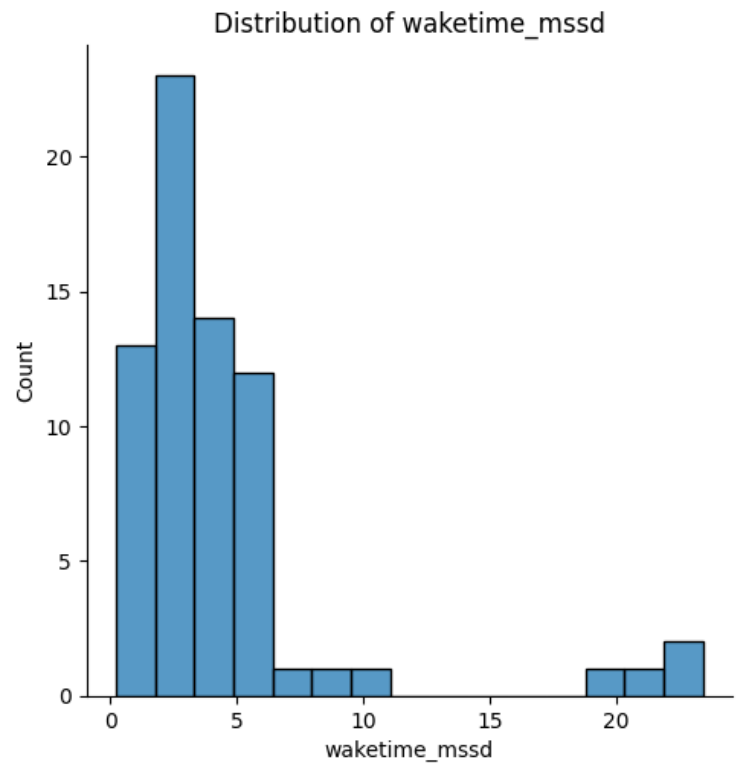
After removing outliers based on the criterion above, we dropped 75 individuals out of 187 individuals. Hence, the final individuals we conducted analysis with were 112 individuals.

Figure 2.1.5 and **Figure 2.1.6** show the distributions of **Figure 2.1.3** and **Figure 2.1.4** after removing outliers for *bedtime_mssd* and *waketime_mssd*.



Left: **Figure 2.1.5**

Right: **Figure 2.1.6**



Then, after running the regression model again, we were able to find statistically significant relationships with sleep features and compliance that we expect to find relationships with.

Results

After running simple linear regression model for each sleep feature against different definition of compliance, we found statistically significant relationships between all three definitions of

compliance with feature are *bedtime*, *bedtime_weekday*, , *waketime_weekday*, *midpoint_sleep*, *midpoint_sleep_weekday*, *midpoint_sleep_weekend*, and *frac_nights_with_data*.

For reference, *bedtime* is mean minutes after 6PM an individual went to sleep, *bedtime_weekday* is mean minutes after 6PM an individual went to sleep on weekdays, *waketime* is mean minutes after an individual woke up after 4AM, *waketime_weekday* is mean minutes after an individual woke up after 4AM on weekdays, *midpoint_sleep* is average of *bedtime* and *waketime* of an individual, indicating mean midpoint of sleep, *midpoint_sleep_weekday* is average of *bedtime_weekday* and *waketime_weekday*, indicating mean midpoint of sleep on weekdays, *midpoint_sleep_weekend* is average of *bedtime_weekend* and *waketime_weekend*, indicating mean midpoint of sleep on weekends, and *frac_nights_with_data* is total nights fitbit data was collected divided by total dates an individual participated in the study.

Figure 2.1.7 - Figure 2.2.8 show features that have significant relationships with all three definitions of compliances - FullCompliance, AttemptCompliance, and StdCompliance.

Figure 2.1.7 - Figure 2.1.9 are regression plots of bedtime and FullCompliance, AttemptCompliance, and StdCompliance. There are statistically significant negative correlations with bedtime and FullCompliance, AttemptCompliance, and StdCompliance. There are also statistically significant negative correlations with bedtime_weekday and FullCompliance, AttemptCompliance, and StdCompliance.

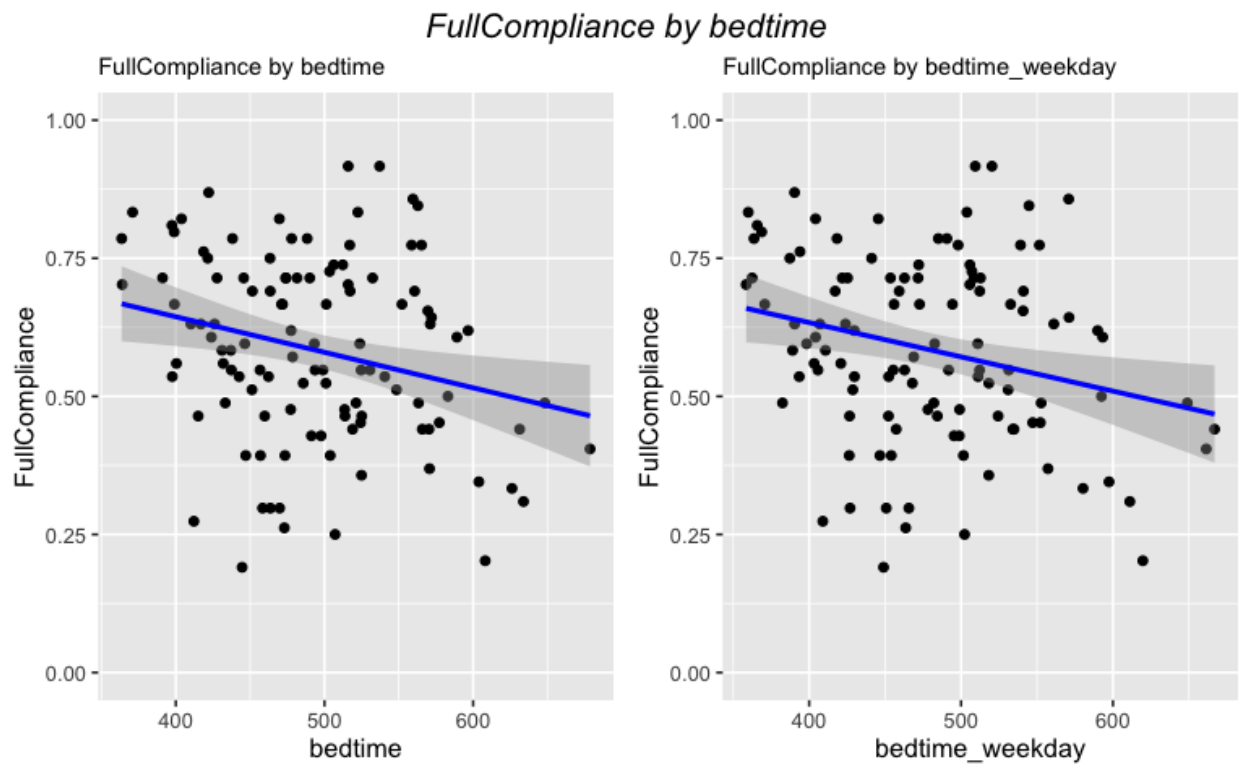


Figure 2.1.7

AttemptCompliance by bedtime

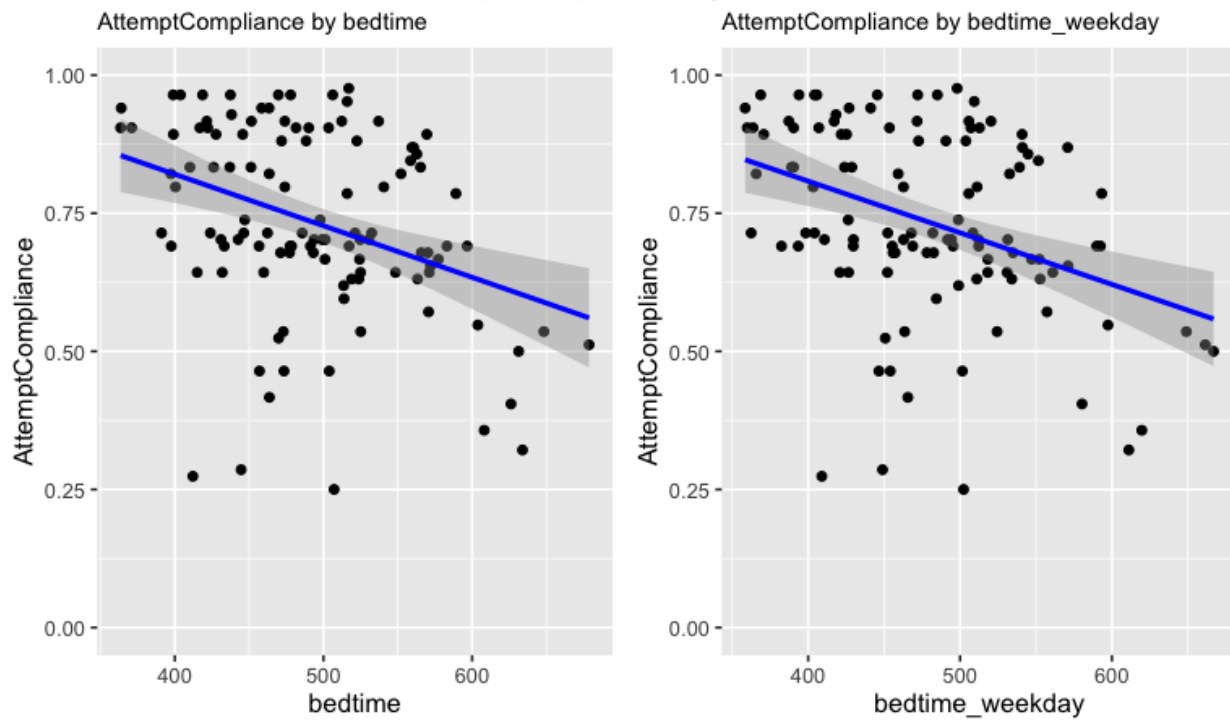


Figure 2.1.8

StdCompliance by bedtime

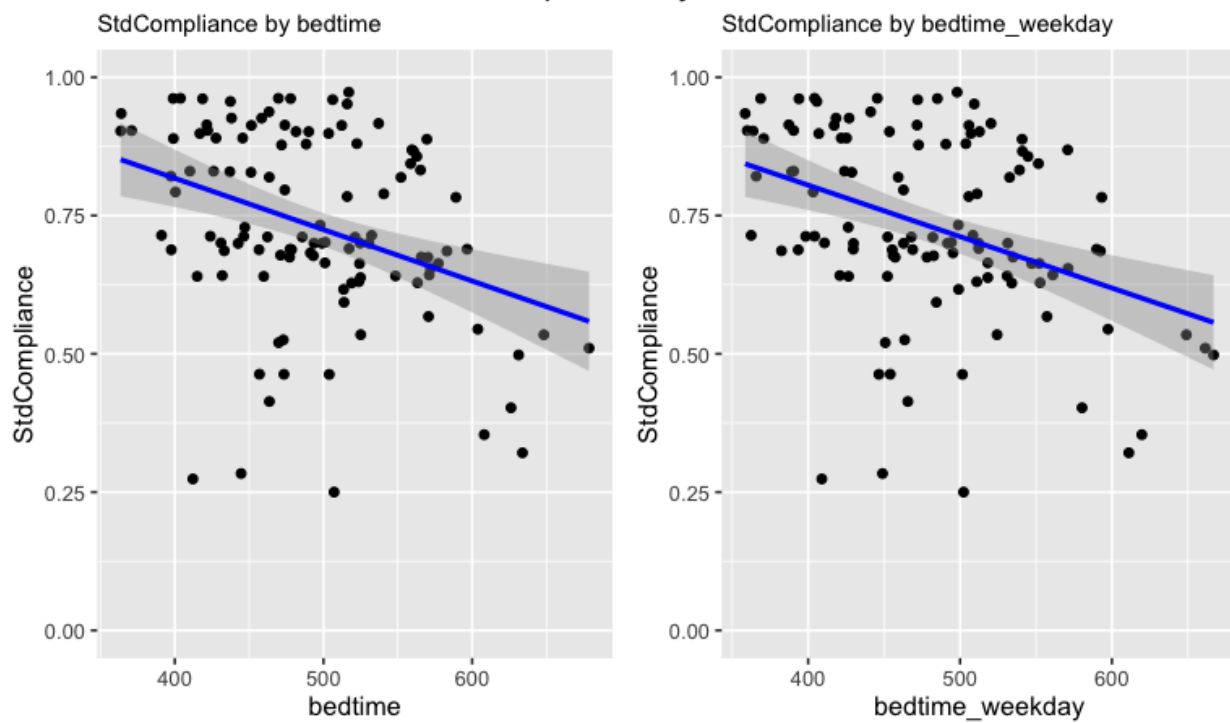


Figure 2.1.9

Figure 2.1.9.1 - Figure 2.1.9.9 are regression plots of waketime and FullCompliance, AttemptCompliance, and StdCompliance. There are statistically significant negative correlations with waketime and FullCompliance, AttemptCompliance, and StdCompliance. There are also statistically significant negative correlations with waketime_weekday and FullCompliance, AttemptCompliance, and StdCompliance.

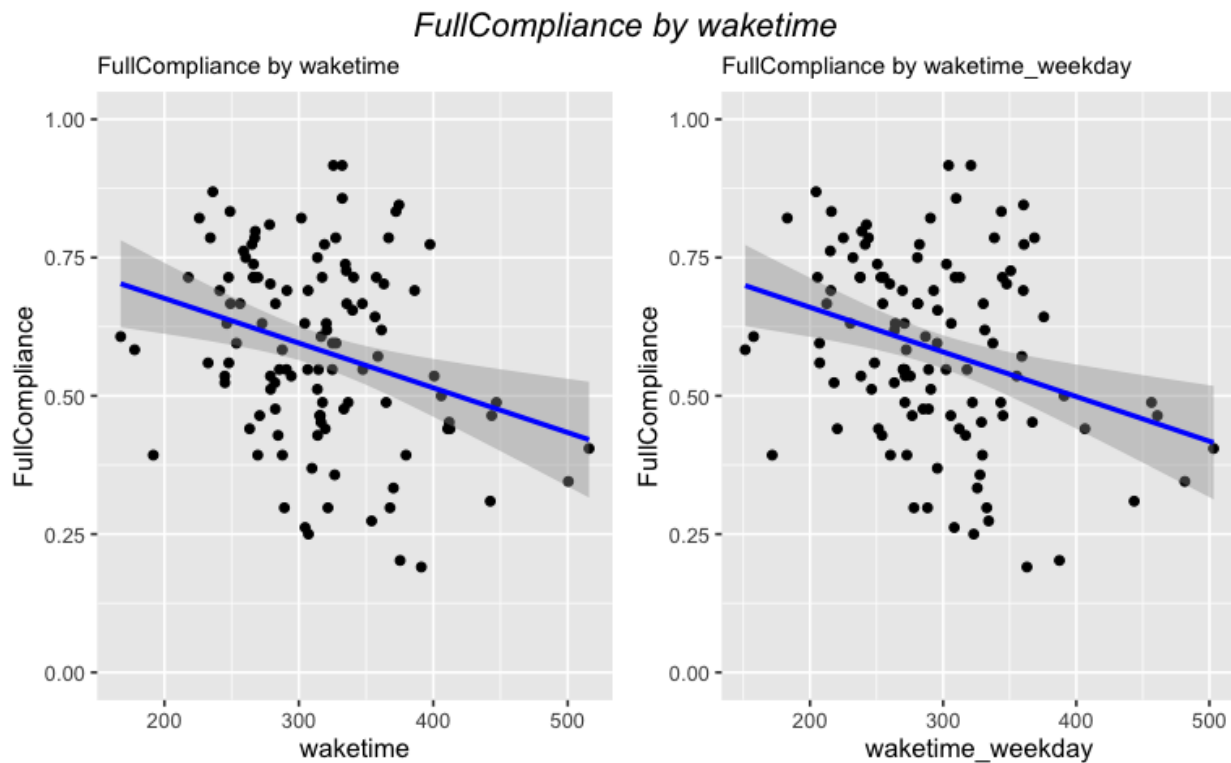


Figure 2.1.9.1

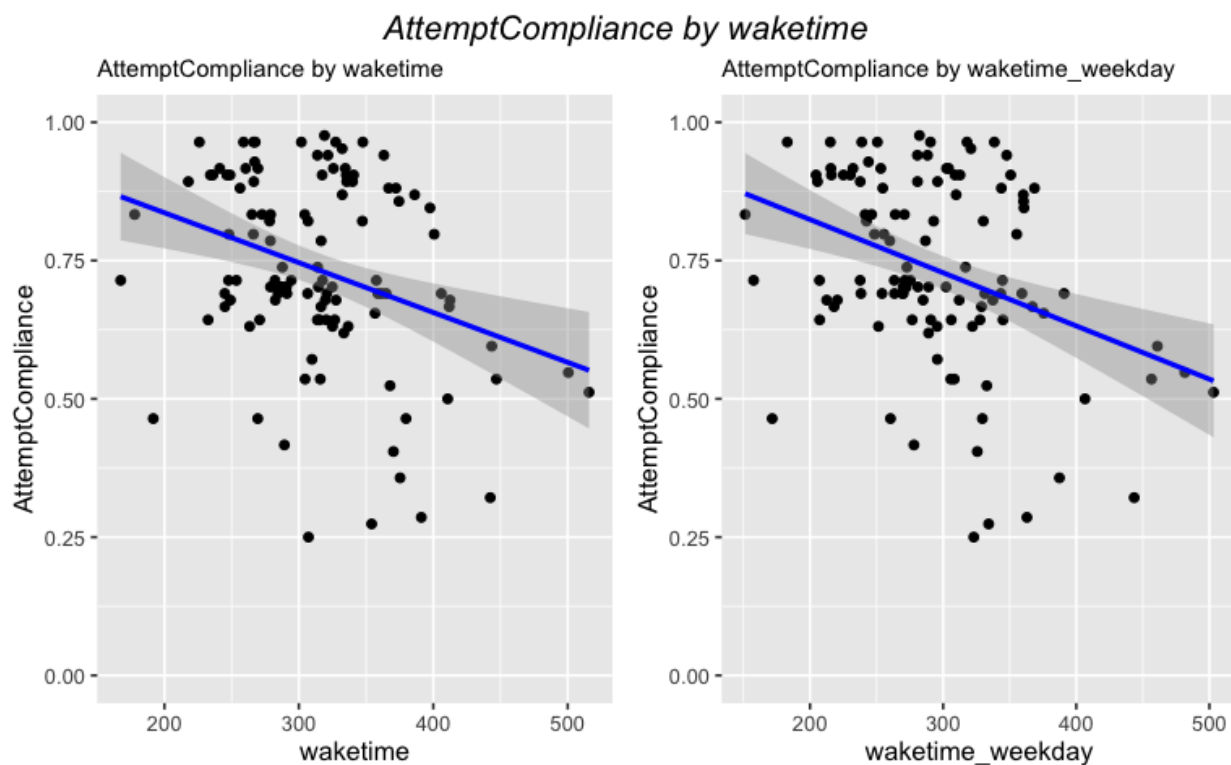


Figure 2.1.9.2

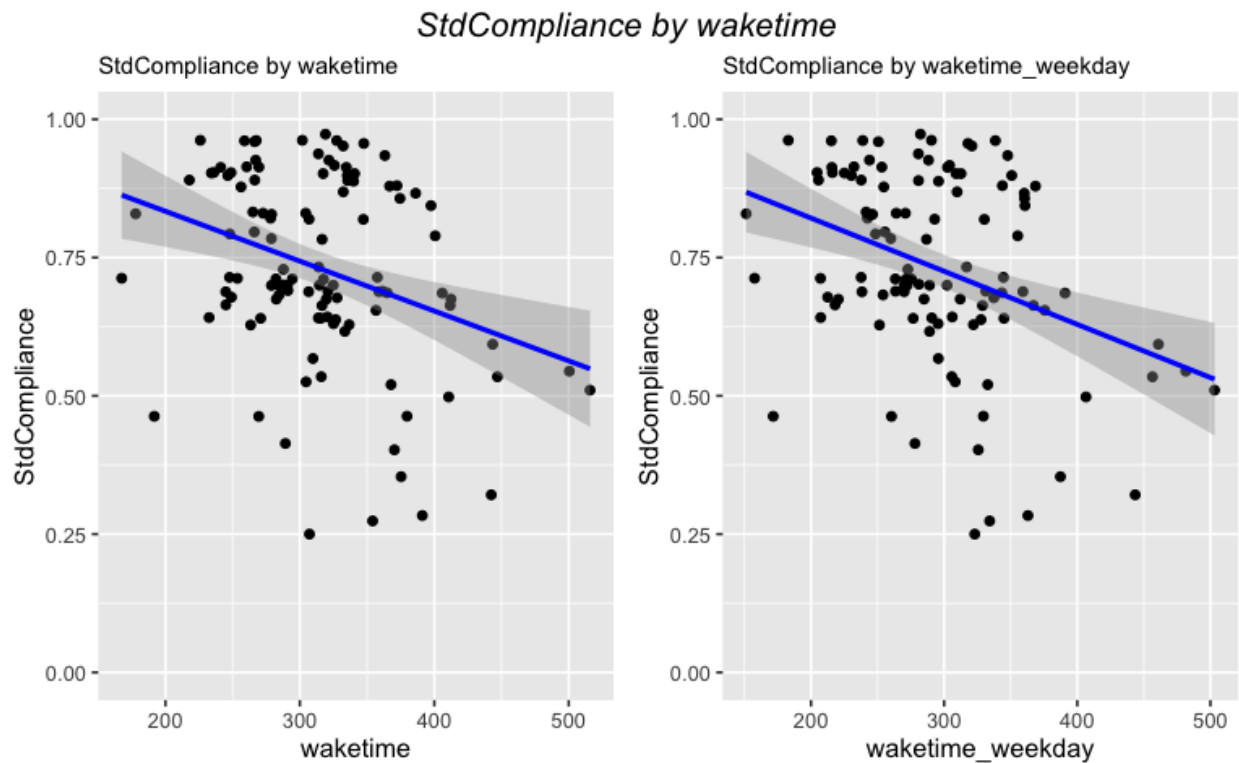


Figure 2.1.9.3

Figure 2.1.9.4 - Figure 2.1.9.6 are regression plots of midpoint_sleep and FullCompliance, AttemptCompliance, and StdCompliance. There are statistically significant negative correlations with midpoint_sleep and FullCompliance, AttemptCompliance, and StdCompliance. There are also statistically significant negative correlations with midpoint_sleep_weekday and FullCompliance, AttemptCompliance, and StdCompliance. Moreover, we see statistically significant negative correlations with midpoint_sleep_weekend and FullCompliance, AttemptCompliance, and StdCompliance.

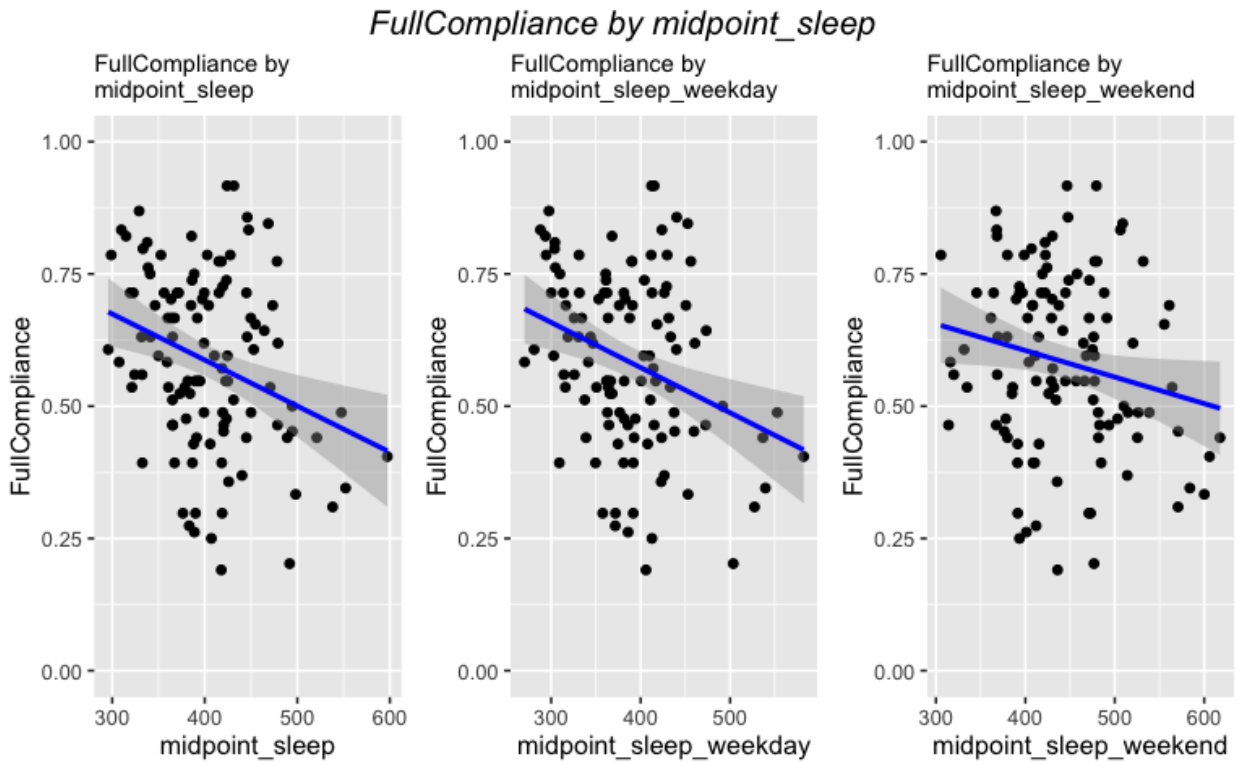


Figure 2.1.9.4

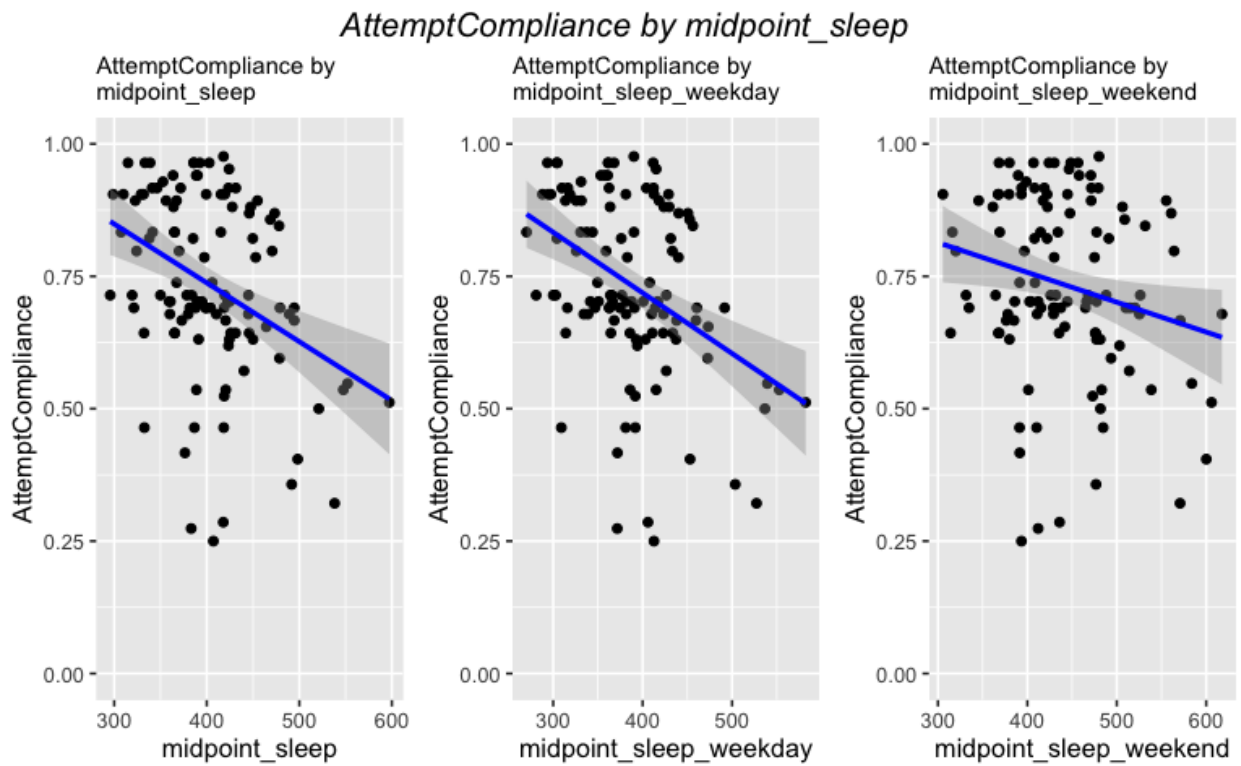


Figure 2.1.9.5

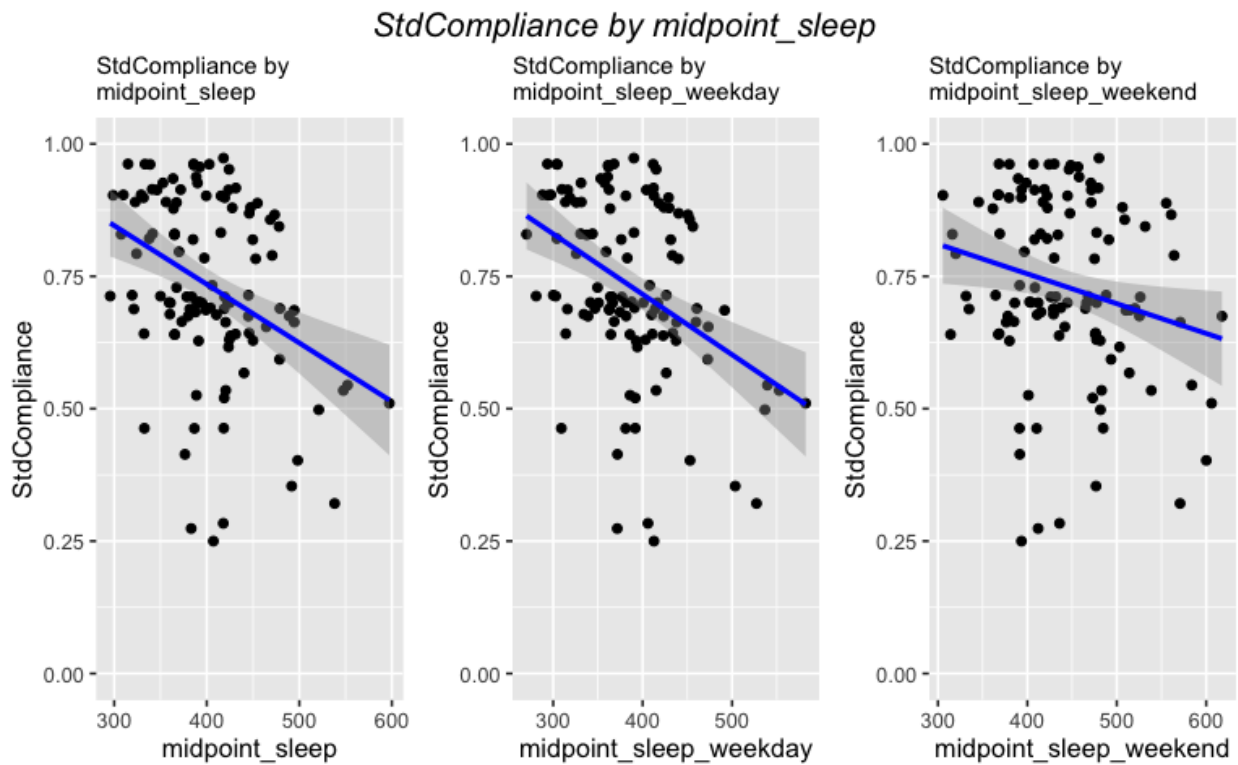
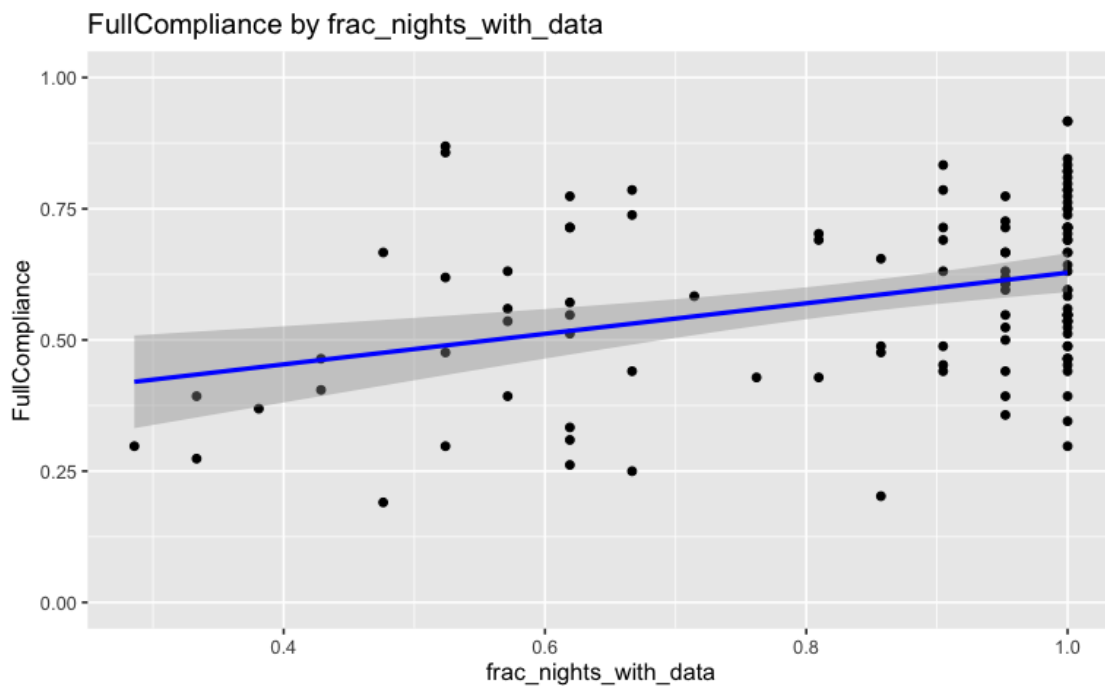


Figure 2.1.9.6

Figure 2.1.9.7 - Figure 2.1.9.9 are regression plots of frac_nights_with_data and FullCompliance, AttemptCompliance, and StdCompliance. There are statistically significant



positive correlations with frac_nights_with_data and FullCompliance, AttemptCompliance, and StdCompliance.

Figure 2.1.9.7

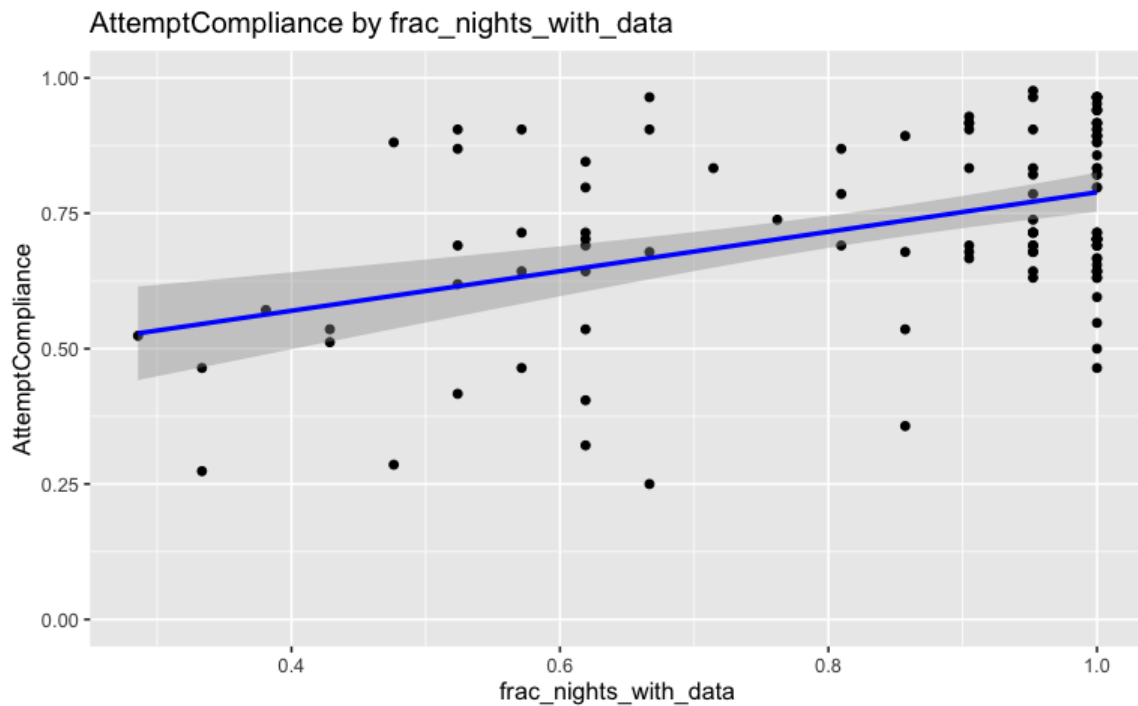


Figure 2.1.9.8

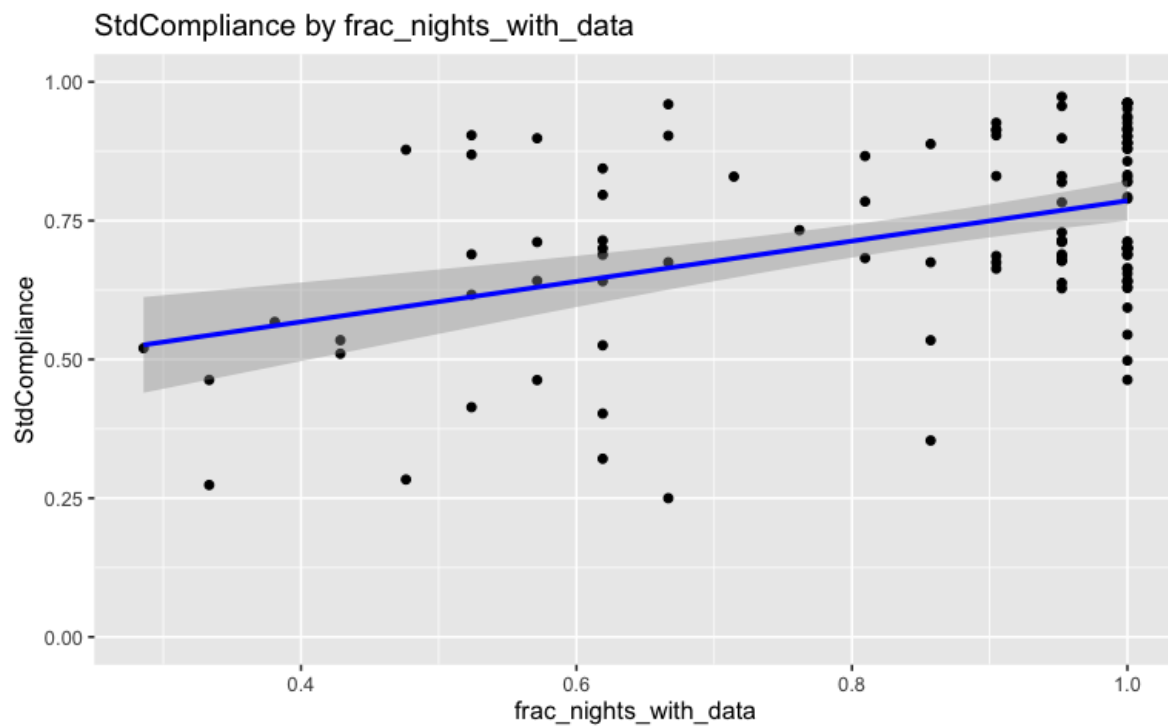


Figure 2.1.9.9

Overall, *bedtime*, *bedtime_weekday*, *waketime*, *waketime_weekday*, *midpoint_sleep*, *midpoint_sleep_weekday*, and *midpoint_sleep_weekend* have statistically significant negative correlations with all definitions of compliances. These results show that individuals who go to bed later on average, who go to bed later on weekdays on average, who wake up earlier on average, who wake up earlier on weekdays on average, who have later midpoint sleep times on average, on weekdays, and on weekends on average have lower compliance.

And *frac_nights_with_data* have statistically significant positive correlations with all three definitions of compliance. This result shows that total nights fitbit data was collected divided by total dates an individual participated in the study are positively correlated with compliance. Hence, we can interpret this result as a sanity check that individuals who wore fitbit to sleep as they were instructed to were also compliant to EMA throughout the studies.

The features that are found to be only significant with FullCompliance are *bedtime_std* and *waketime_mssd*. *bedtime_std* is the standard deviation of *bedtime*, which is the mean minutes after an individual went to sleep after 6PM. *waketime_mssd* stands for mean successive squared difference of waketime. Mean successive squared difference measures variability of successive items. For example, if waketime for an individual is [100, 122, 111] = waketime_mssd is $((122 - 100)^2 + (111 - 122)^2) / 2 = 302.5$. We found statistically significant negative correlations between FullCompliance and *bedtime_std* and FullCompliance and *waketime_mssd*. This results shows that individuals with higher variations of bedtime and variations of successive waketime have lower FullCompliance.

Figure 2.1.9.9.1 - Figure 2.1.9.9.2 are regression plots of FullCompliance by *bedtime_std* and FullCompliance by *waketime_mssd*.

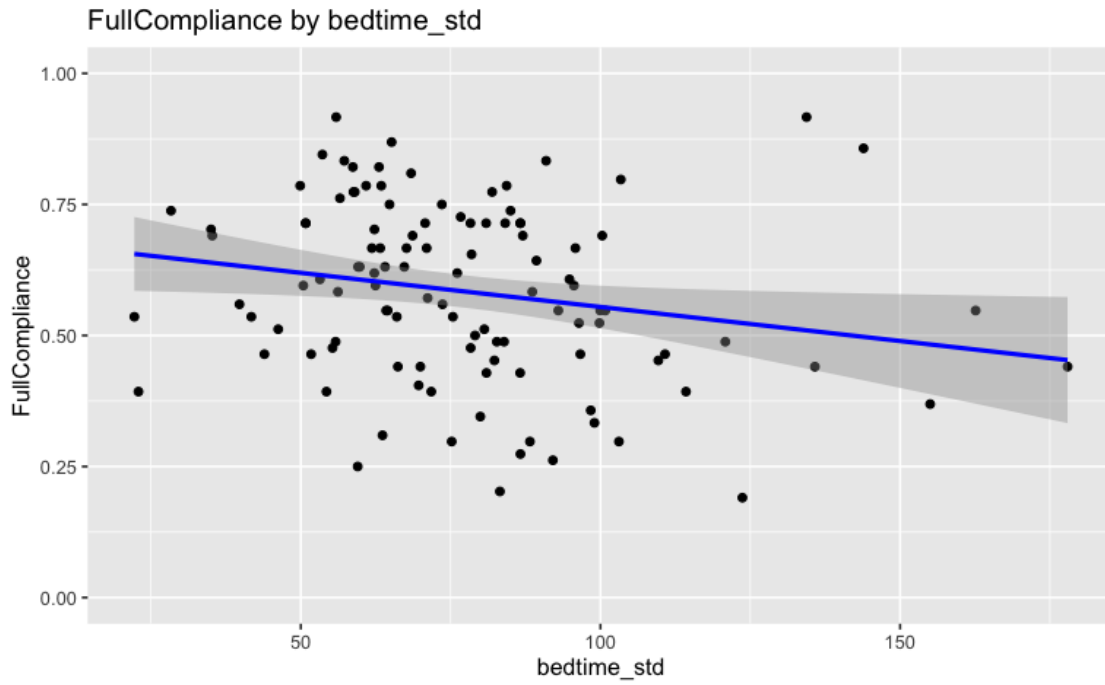


Figure 2.1.9.9.1

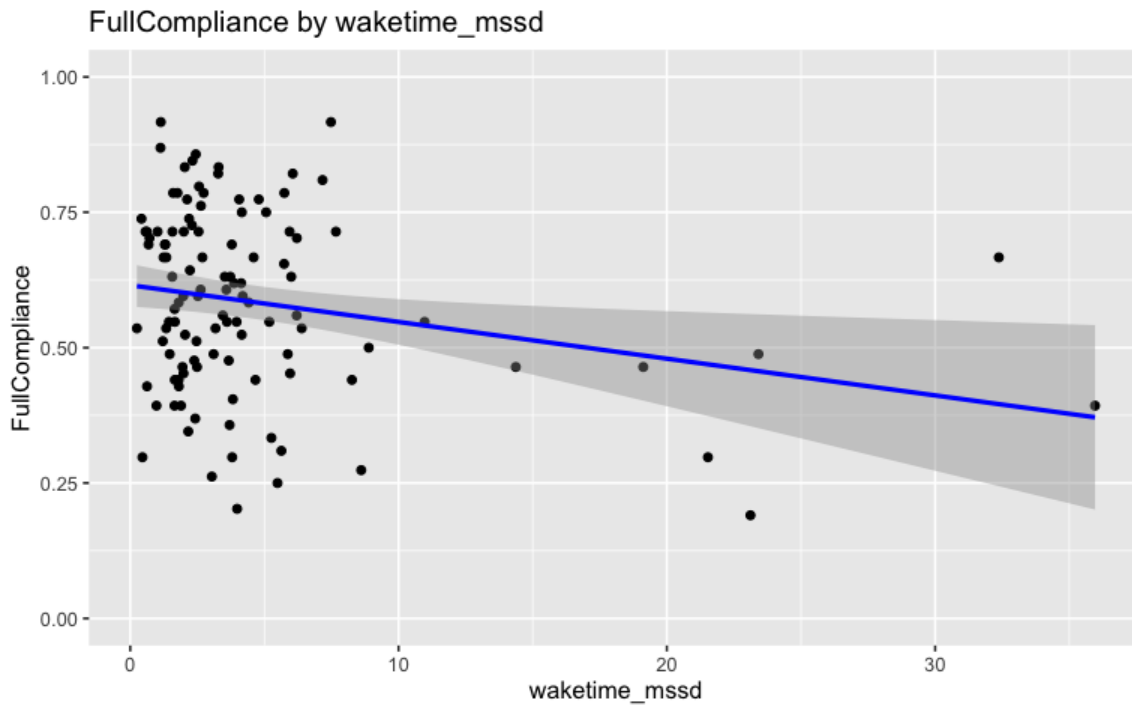


Figure 2.1.9.9.2

The features that are found to be only significant with AttemptCompliance and StdCompliance are *avg_nap_length* and *bedtime_weekend*.

avg_nap_length is the mean length of naps an individual took, *bedtime_weekend* is the mean number of minutes after 6PM an individual went to sleep on weekends. We found statistically significant negative correlations with both AttemptCompliance and StdCompliance with *avg_nap_length*. We also found statistically significant negative correlations with both AttemptCompliance and StdCompliance with *bedtime_weekend*. From this result, we can claim that individuals who take a higher number of naps have lower AttemptCompliance and StdCompliance. Moreover, individuals who go to bedtime later on weekends also have lower AttemptCompliance and StdCompliance.

Figure 2.1.9.9.3 - Figure 2.1.9.9.4 are regression plots of AttemptCompliance by *avg_nap_length* and StdCompliance by *avg_nap_length*.

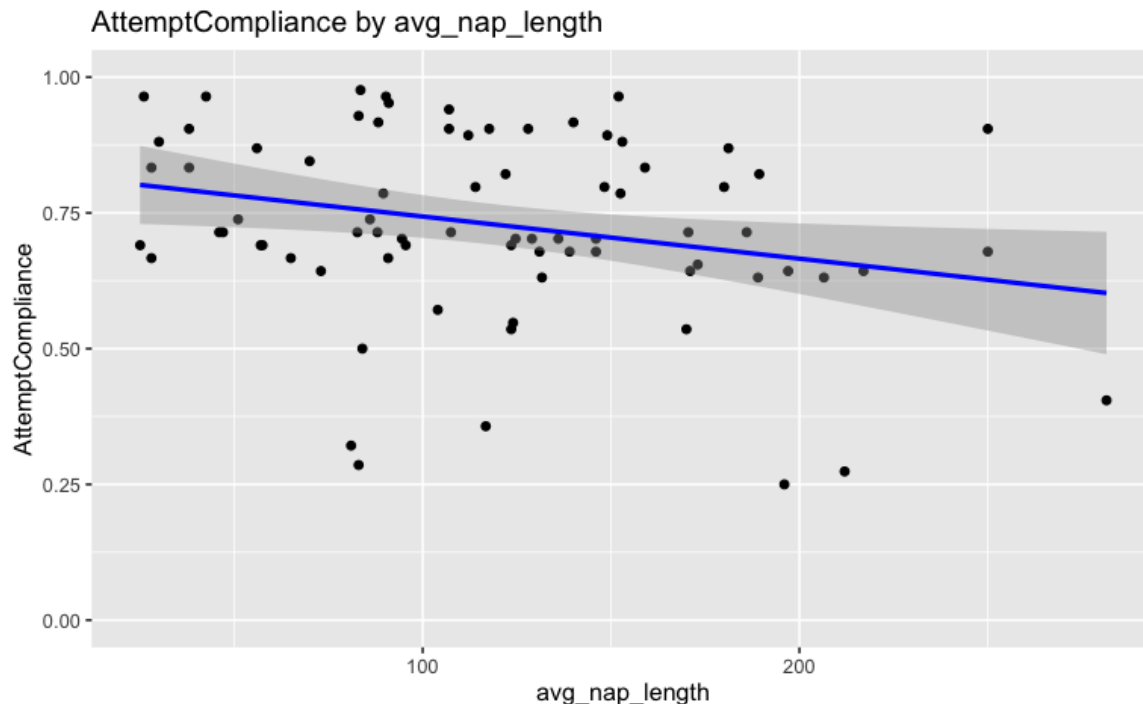


Figure 2.1.9.9.3

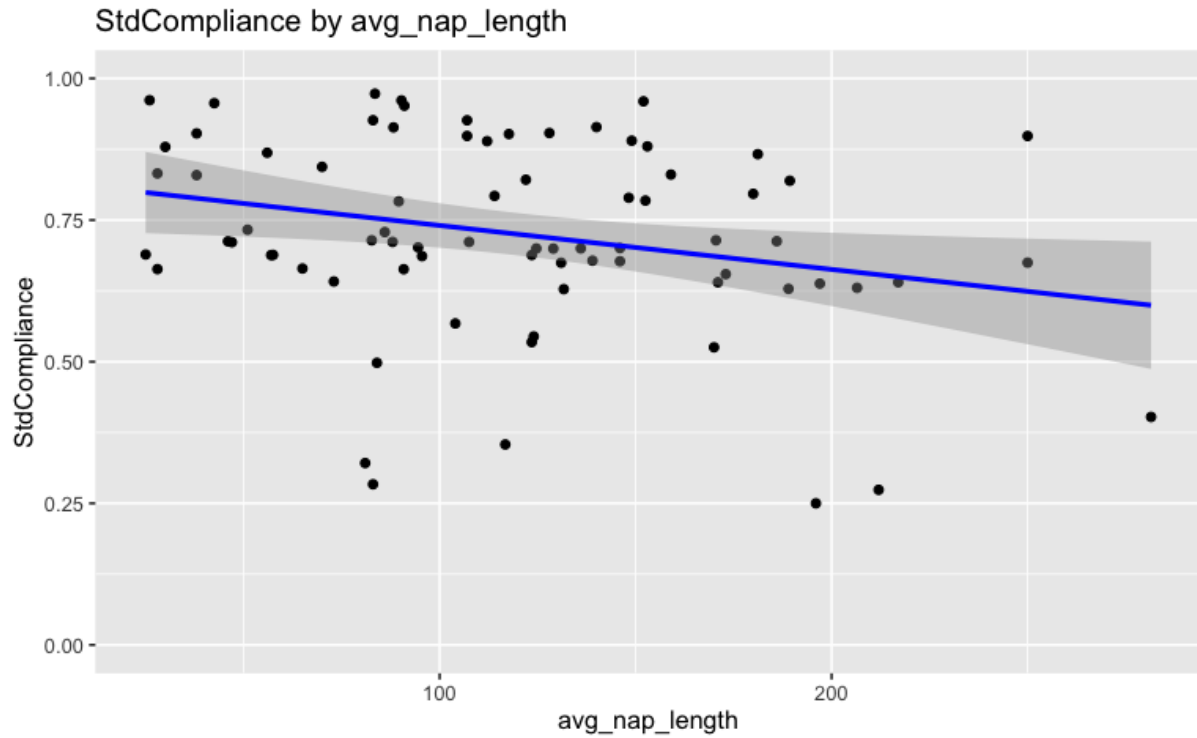


Figure 2.1.9.4

Figure 2.1.9.5 - Figure 2.1.9.6 are regression plots of AttemptCompliance by *bedtime_weekend* and StdCompliance by *bedtime_weekend*.

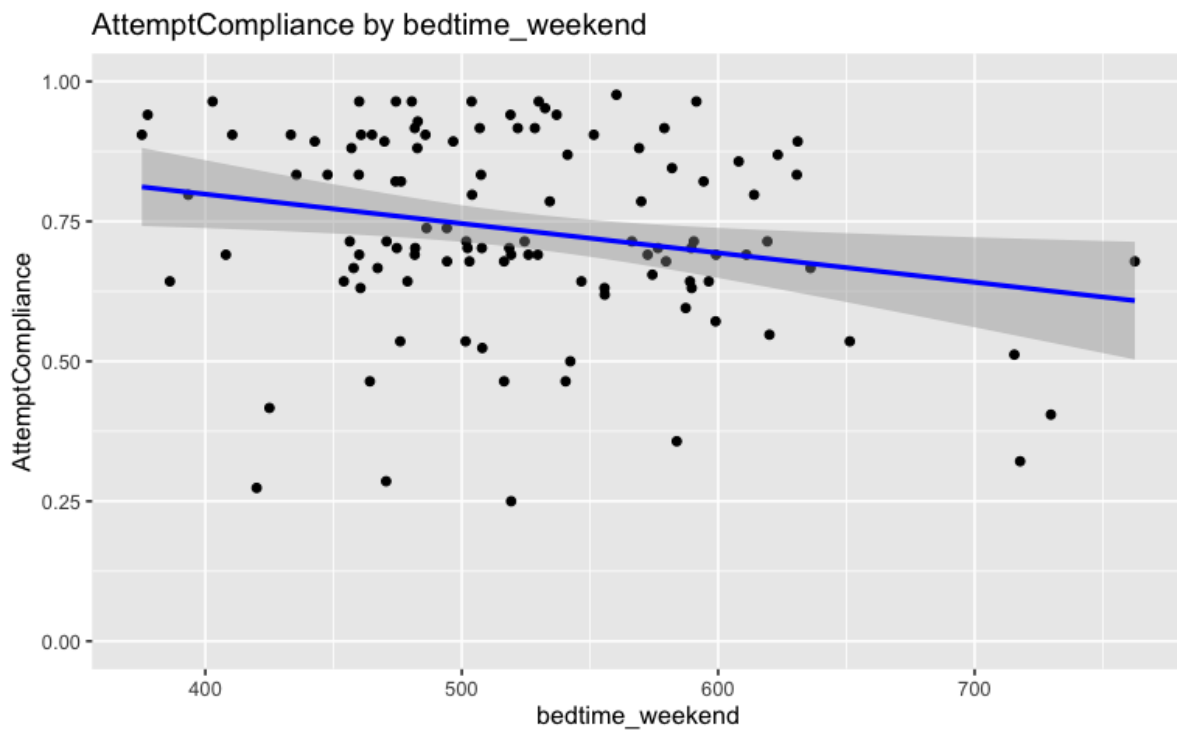


Figure 2.1.9.9.5

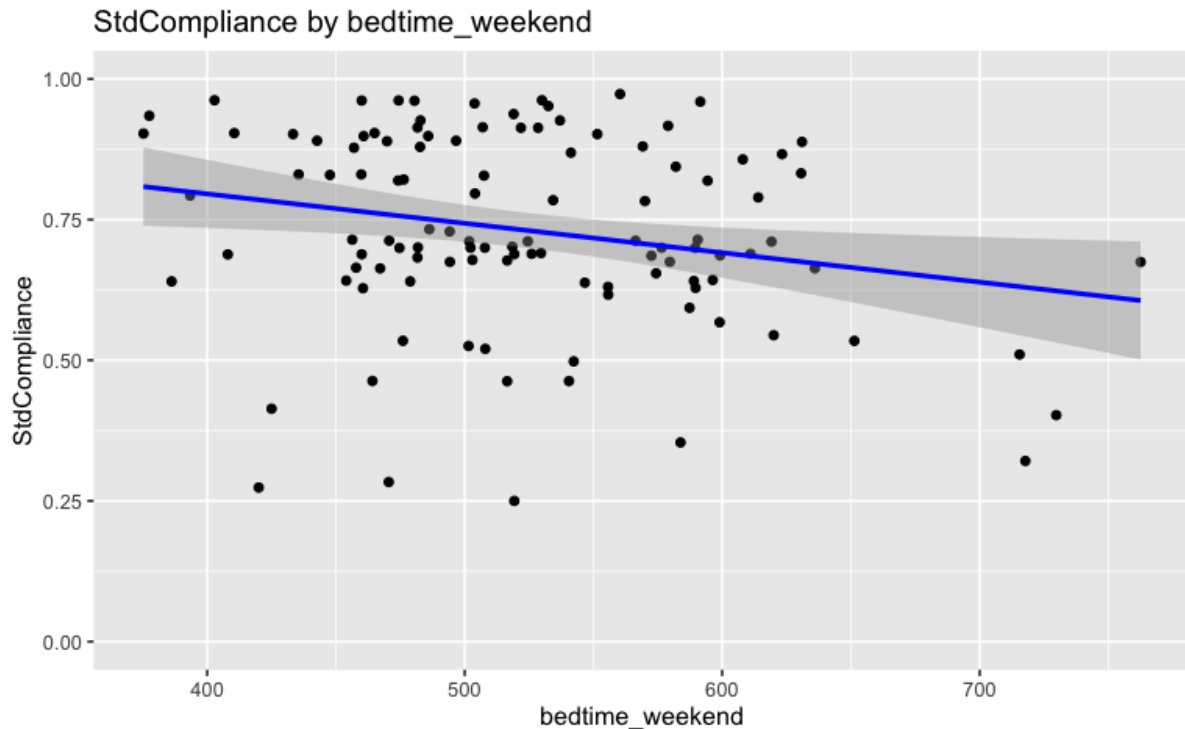


Figure 2.1.9.9.6

To summarize, the results found above, *bedtime*, *bedtime_weekday*, *waketime*, *waketime_weekday*, *midpoint_sleep*, *midpoint_sleep_weekday*, and *midpoint_sleep_weekend* have statistically significant negative correlations with all definitions of compliances. Hence, we know that individuals who go to bed later on average, who go to bed later on weekdays on average, who wake up earlier on average, who wake up earlier on weekdays on average, who have later midpoint sleep times on average, on weekdays, and on weekends on average have lower compliance.

And *frac_nights_with_data* have statistically significant positive correlations with all three definitions of compliance. This result shows that total nights fitbit data was collected divided by total dates an individual participated in the study are positively correlated with compliance. Hence, we can interpret this result as a sanity check that individuals who wore fitbit to sleep as they were instructed to were also compliant to EMA throughout the studies.

Moreover, we found statistically significant negative correlations between FullCompliance and *bedtime_std* and FullCompliance and *waketime_mssd*. This results shows that individuals with higher variations of bedtime and variations of successive waketime have lower FullCompliance.

Furthermore, we found statistically significant negative correlations with both AttemptCompliance and StdCompliance with *avg_nap_length*. We also found statistically significant negative correlations with both AttemptCompliance and StdCompliance with *bedtime_weekend*. From this result, we can claim that individuals who take a higher number of naps have lower AttemptCompliance and StdCompliance. Moreover, individuals who go to bedtime later on weekends also have lower AttemptCompliance and StdCompliance.

- Thus, with fitbit sleep data, we found significant relationships between time individuals go to bed, time individuals wake up, and midpoint of time individuals are sleeping, and possibly variation of time individuals go to bed, variation of time individuals wake up, and Length of naptime

2.2 *Smartphone Data and Compliance.*

Abstract

Smartphones have become an inevitable part of individuals' lives. Hence, understanding smartphone usages elucidates distinct characteristics of individuals. With smartphone usages collected from an open-source context instrumentation framework called AWARE, we investigated the relationships between phone usages and all three definitions of compliance (Attempt, Standard, and Full). We found no significant relationships between phone usage and compliance. The result suggests that smartphones and compliance have no correlation. Hence, smartphones could be a good medium for monitoring EMA.

Introduction

The Life@CMU Phase 1 study collected smartphone data of individuals who participated in the study with an open-source context instrumentation framework called AWARE. Data was collected locally on individuals' phones, and the AWARE application collected phone usages of individuals as well as bluetooth addresses, call logs, and locations. In our analysis, we conducted analysis with phone usages of individuals and aimed to find relationships between phone usage and compliance. For each hour, number of screen-on events, number of unlock events, average time between unlock and off events were recorded.

Method

Smartphone data consists of three smartphone features *mean_screen*, *mean_unlocks*, and *mean_minutes*. We computed the average of each feature over three weeks of Life@CMU Phase 1 study for each individual. And we computed the average of each definition of compliance (Full, Attempt, and Standard) over three weeks of Life@CMU Phase 1 study.

We then joined the mean smartphone feature over three weeks and mean compliance of Full, Attempt, and Standard on participant ID. And we built simple linear regression models of each definition of compliance (Full, Attempt, and Standard) on each feature of the dataset.

mean_screen is the average number of screen-on events over the week (averaging the hourly counts), *mean_unlocks* is the average number of unlock events over the week (averaging the hourly counts), and *mean_minutes* is average number of time spent on one's phone over the week (averaging the hourly means).

Results

We did not find any features that had a statistically significant relationship with three definitions of compliance. This result may be due to the fact that the features are averaged across a three weeks period. Hence, we might not be capturing compliance with mean phone usages. Overall, this result shows that smartphones may be a good medium for monitoring EMA surveys, as we found that smartphones and compliance are not correlated.

3 Contextual Features & Compliance.

3.1 Survey Quality and Compliance.

Abstract

Straightlining is a satisficing behavior that indirectly measures quality of survey. In our analysis, we computed straightlining using a metric called *scale point variation* of a battery of questions that was asked four times a day, where a battery of questions is defined as a group of questions that are in the same question category, that have the same scale points, and the same response variables. The battery of questions that meets the criterion is Affect questions, which asks for how individuals are feeling at the moment the question is asked. After calculating scale point variation, we investigated the relationships between scale point variation and all three definitions of compliance (Attempt, Standard, and Full). We found statistically significant negative relationships between standard deviation of *scale point variation* and Full, Attempt compliance, and Standard compliance. The results suggest that variations of survey quality could be a predictor of compliance.

Introduction

Behaviors such as selecting the first response option that seems reasonable, agreeing with assertions, endorsing the status quo, giving “don’t know” or “not applicable” responses, randomly choosing answers, and failing to differentiate between response alternatives (i.e., straightlining), are different strategies of *satisficing*. (Krosnick, 1991).

Previous literature has examined what types of people are more likely to satisfice in surveys and the effects of satisficing behaviors on reliability and validity measures of surveys, but no literature in the past has looked at the relationship between satisficing and compliance. In this paper, we used the theory of satisficing to evaluate the quality of EMA responses in Life@CMU Phase 1. And using the theory of satisficing, we examined the relationship between survey quality and compliance.

Among different behaviors of satisficing, we specifically investigated *straightlining*, a behavior defined as “when respondents fail to differentiate between the items with their answers by giving identical (or nearly identical) responses to all items using the same response scale (Yan, 2008).” We calculated straightlining using the metric called *scale point variation* of a battery of questions that was asked four times a day in the beginning of every survey throughout the studies (Affect) and computed the standard deviation of scale point variation score over all days for each individual who answered the Affect questions. Then, we built a simple linear regression model of mean compliance over 21 days on standard deviation of mean scale point variation score over 21 days.

Method

Past study by Linville, Salovey, and Fischer (1986) practiced the probability of differentiation, called scale point variation. This metric measures variation of responses, and it aims to capture straightlining. The metric is defined as $1 - \sum p_i^2$, where p_i is the proportion of the values rated at each scale point on a rating scale and i indicates the number of scale points” (Kim et al., 2019). The value ranges from 0 to a value close to 1, 0 indicating most straightlining and a value close to 1 indicating least straightlining. For example, if an individual responded “Extremely”, for all seven questions in one battery of questions, where a battery of questions is defined as a group of questions that are in the same question category, that have the same scale points, and the same response variables, the individual would have $(1 - (7/7)^2) = 0$. But if an individual responded “Not at all” for three questions and “Extremely” for four questions within the same battery of questions, the individual would have $(1 - \{ (3/7)^2 + (4/7)^2 \}) = 0.49$. Using this metric, we calculated the straightlining score of individuals on a battery of questions that are asked four times a day and consists of more than two questions in order to make sure straightlining is fully captured within a battery of questions.

The battery of questions that meets the criteria defined above in Life@CMU data is Affect questions, that ask how individuals are feeling at the moment. The question has 5 scale points and the battery of questions consists of seven questions, asking for how anxious, depressed, frustrated, happy, overwhelmed, lonely, and connected to others an individual is feeling at the moment. Ranging from 0 to 5, the scale points represent an increasing level of intensity, 0 indicating “Not at all” and 4 indicating “Extremely.” We calculated scale point variation of the Affect battery of questions for each day and computed **standard deviation of scale point variation over 21 days** in Phase1.

Figure 3.1.1 below shows an example of a schematic of calculation with Life@CMU Phase1 for an individual with ID 102 who filled out the seven Affect questions on Day 1 Time 1. Because

the individual answered three 0s, two 1s, and two 2s among seven questions, the scale point variation score for Day 1 Time 1 is ($1 - \{ (3/7)^2 + (2/7)^2 + (2/7)^2 \} = 0.6531$).

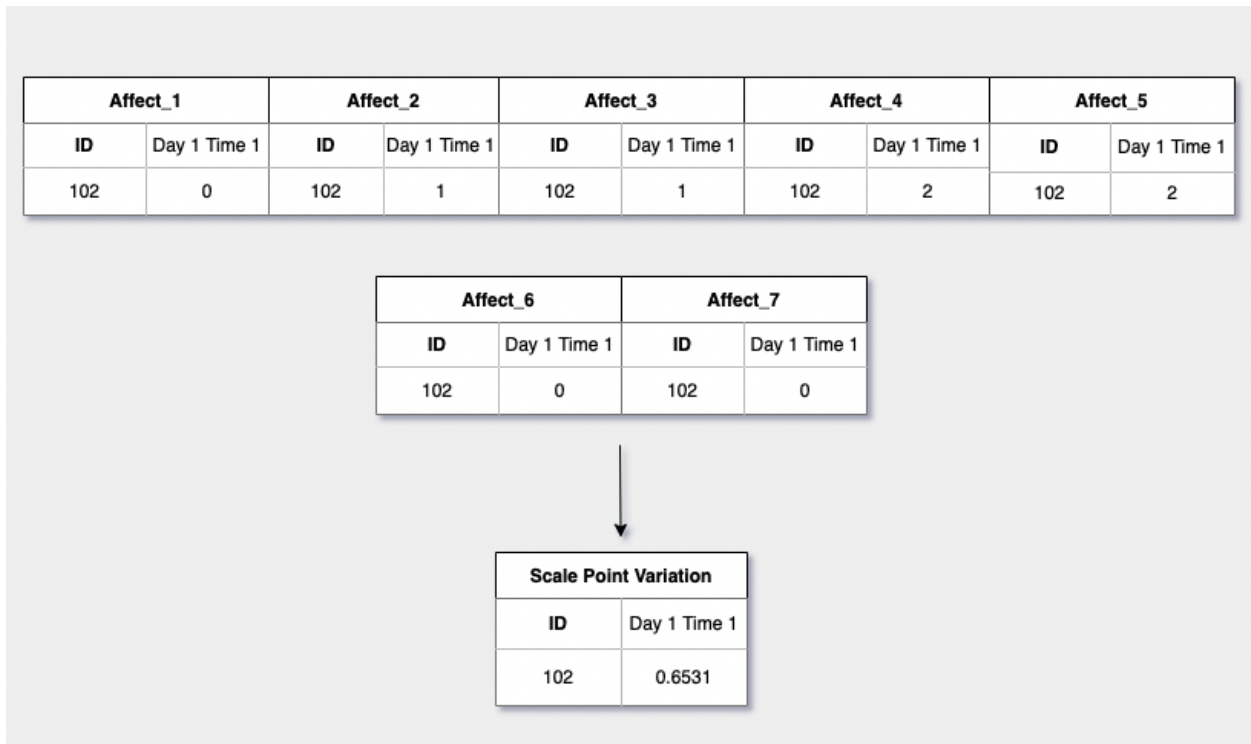


Figure 3.1.1

Figure 3.1.2 below shows an example of a schematic of calculation with Life@CMU Phase1 for an individual with ID 102 who partially filled out the Affect questions on Day 1 Time 1. Because the individual answered only four questions out of seven questions and the individual answered three 0s and one 1, the scale point variation score for Day 1 Time 1 is ($1 - \{ (3/4)^2 + (1/4)^2 \} = 0.3750$). But if the individual did not answer any of the seven questions, the final scale point variation score for Day 1 Time 1 is NaN.

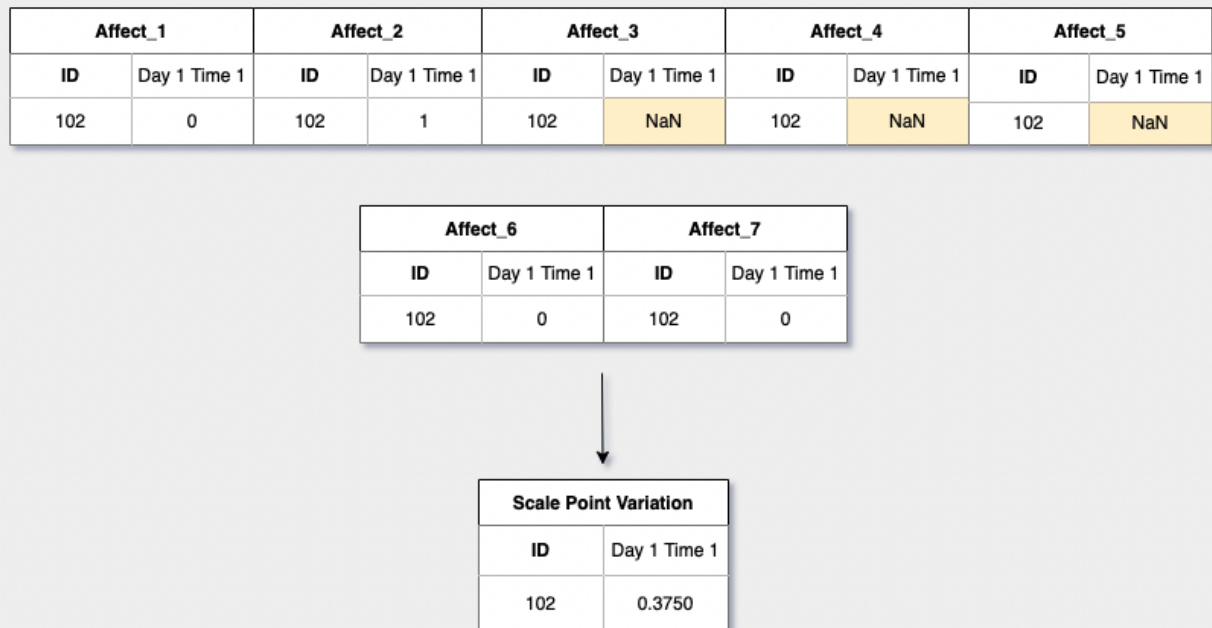


Figure 3.1.2

Figure 3.1.3 below shows an example of a schematic of calculation of scale point variation for an individual with ID 102 for Day 1. We take the mean of four time points. Because the individual's scale point variation score for Time 1 is 0.6531, 0.1242 for Time 2, 0.9312 for Time 3, 0.2412 for Time 4, the mean scale point variation for Day 1 is $(0.6531 + 0.1242 + 0.9312 + 0.2412) / 4 = 0.4874$.

Scale Point Variation		Scale Point Variation		Scale Point Variation		Scale Point Variation	
ID	Day 1 Time 1	ID	Day 1 Time 2	ID	Day 1 Time 3	ID	Day 1 Time 4
102	0.6531	102	0.1242	102	0.9312	102	0.2412



Scale Point Variation	
ID	Day 1
102	0.4874

Figure 3.1.3

Figure 3.1.4 shows an example of a schematic of calculation of scale point variation for an individual with ID 102 who partially answered the Affect questions on Day 1. Because the scale point variation score for Time 1 is 0.6531, Time 2 is 0.1242, and we disregard the NaN values, the scale point variation score for Day 1 is $(0.6531 + 0.1242) / 2 = 0.3887$.

Scale Point Variation		Scale Point Variation		Scale Point Variation		Scale Point Variation	
ID	Day 1 Time 1	ID	Day 1 Time 2	ID	Day 1 Time 3	ID	Day 1 Time 4
102	0.6531	102	0.1242	102	NaN	102	NaN



Scale Point Variation	
ID	Day 1
102	0.3887

Figure 3.1.4

After calculating scale point variation for each day for all 188 individuals, we dropped individuals who had at least one NaN scale point variation score among 21 days, meaning that the individual did not answer Affect questions at all on at least one day. 18 individuals out of 188 individuals did not answer Affect questions at all on at least one day over 21 days. So the final individuals that we conducted analysis with were 170 individuals total. Then, we aggregated the scale point variation score for each individual over 21 days and computed the **standard deviation of scale point variation** over 21 days.

Figure 3.1.5 below shows the final scale point variation score for the individual with ID 102 over 21 days for Life@CMU Phase 1 data.

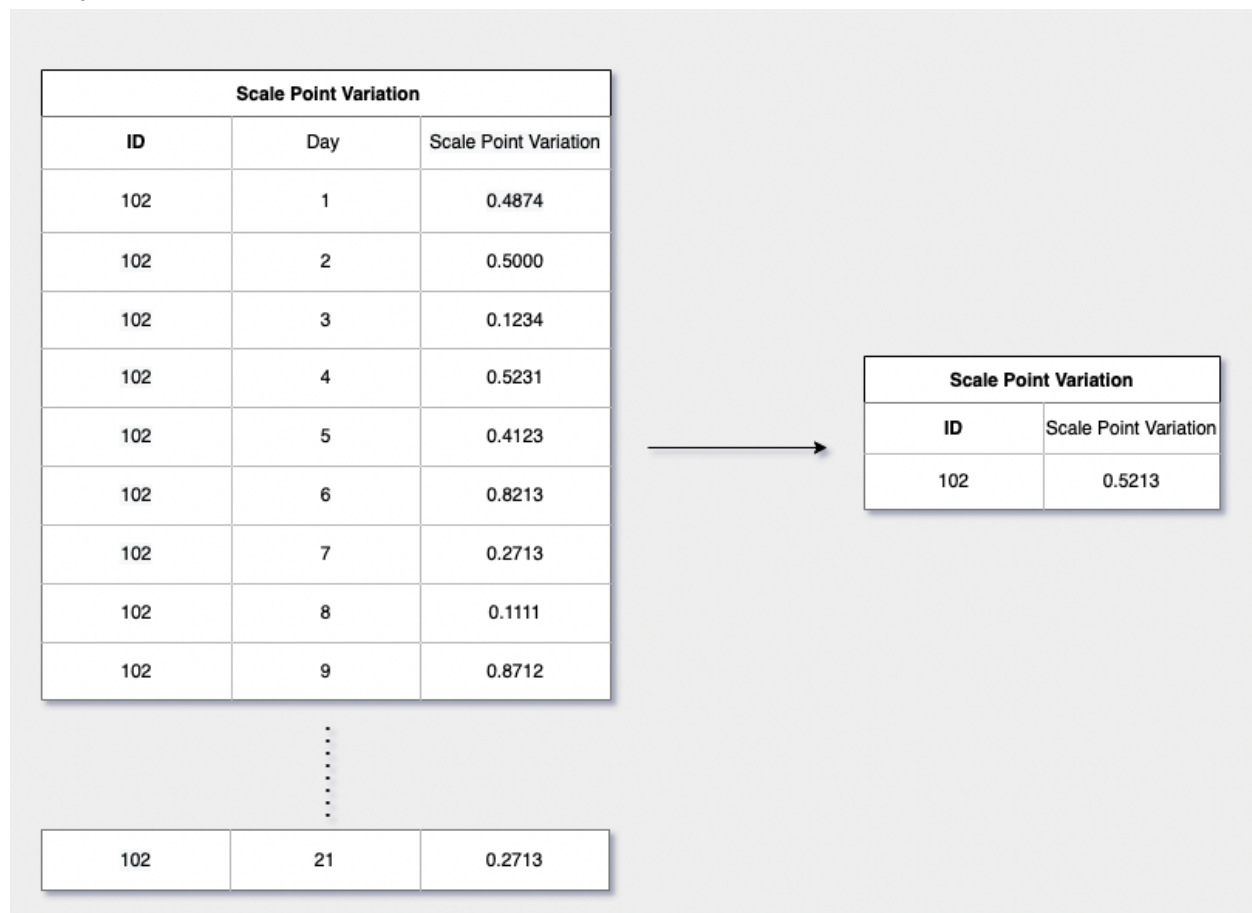


Figure 3.1.5

After we computed the **standard deviation of scale point variation** over 21 days for 170 individuals, we computed the mean of different definitions of compliances (Full, Attempt, and Standard) over 21 days. Then, we built a linear regression model of different definitions of compliances against the scale point variation score for 170 individuals.

Results

Figure 3.1.6 below shows the distribution of standard deviation of scale point variation of 170 individuals. We see a bimodal distribution with the group on the left with low scale point variation and the group on the right with high scale point variation.

Distribution of Standard Deviation of Scale Point Variation Over 21 Days

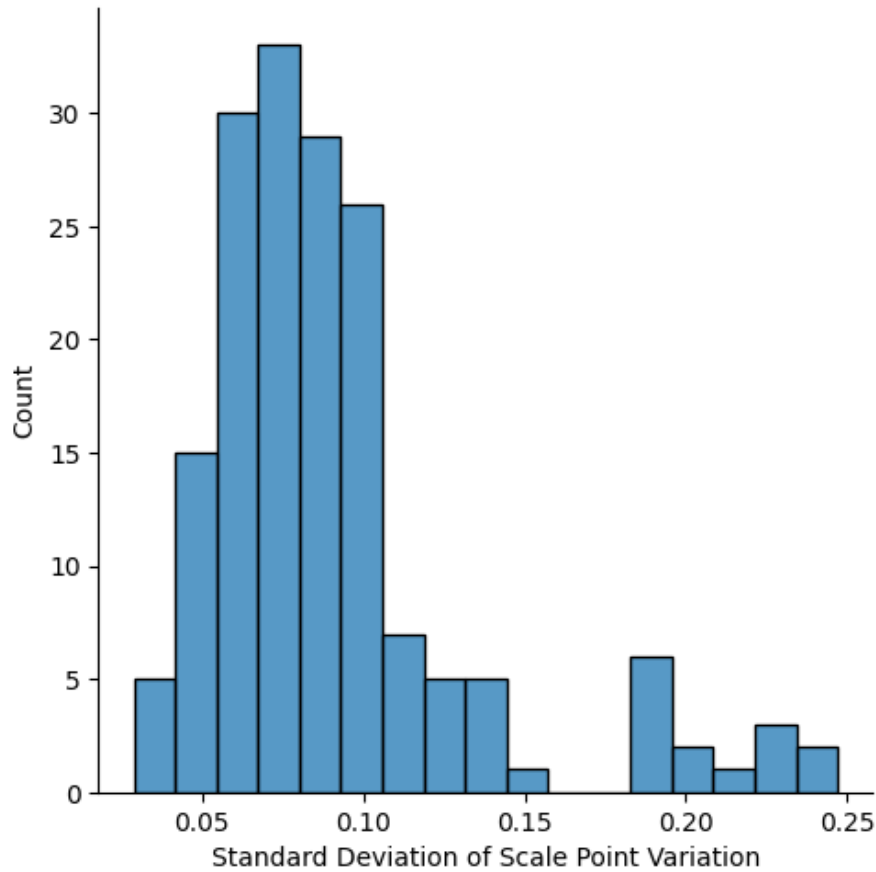


Figure 3.1.6

Figure 3.1.7 below is a regression plot of Full compliance by standard deviation of scale point variation. The p-value associated with scale point variation is 0.0236. Hence, there is a significant statistical negative relationship between Full compliance and standard deviation of scale point variation.

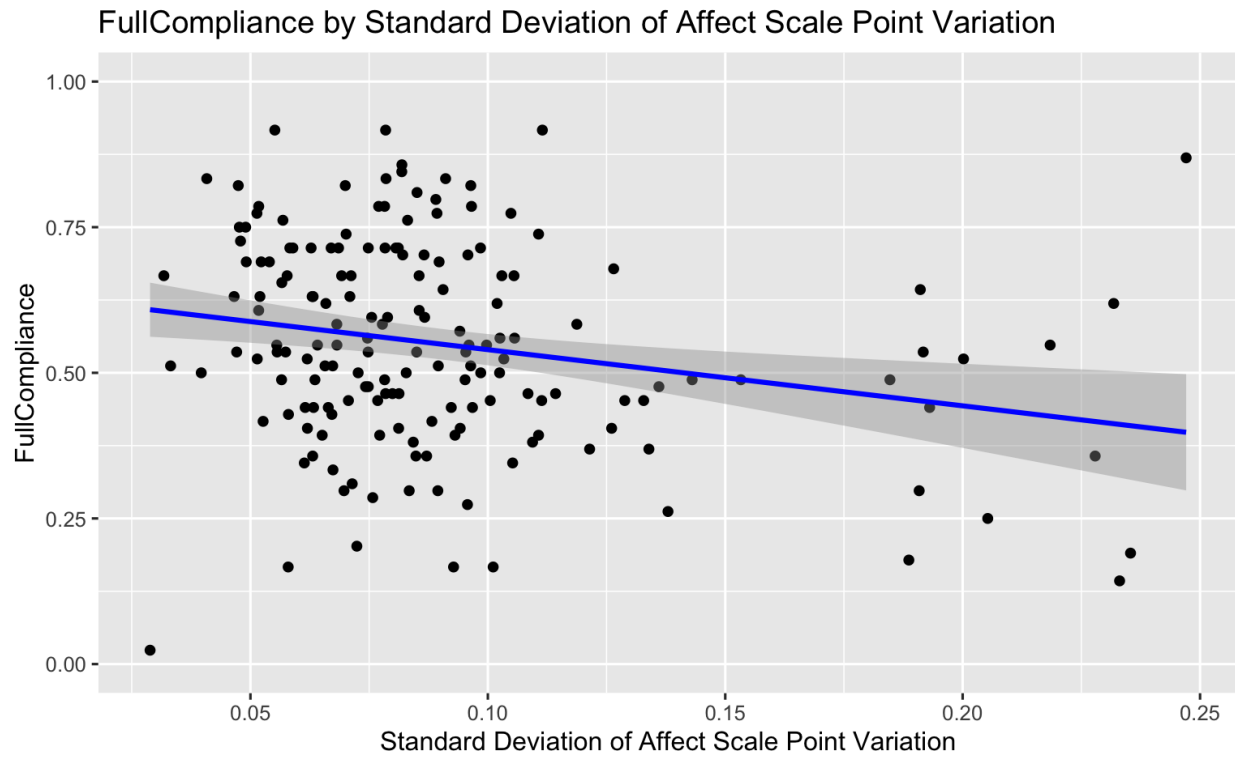


Figure 3.1.7

Figure 3.1.8 below is a regression plot of Attempt compliance by standard deviation of scale point variation. The p-value associated with scale point variation is 0.0000459. Thus, there is statistically significant negative correlation between Attempt compliance and standard deviation of scale point variation.

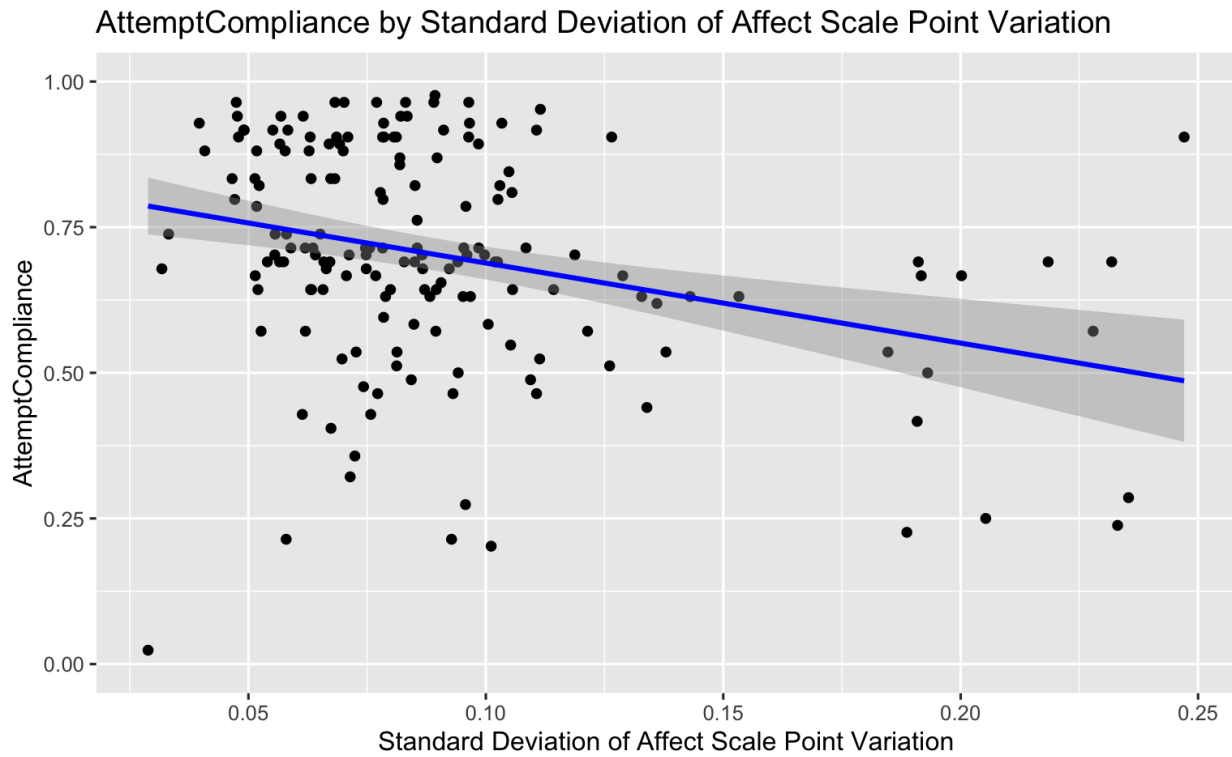


Figure 3.1.8

Figure 3.1.9 below is a regression plot of Standard Compliance by standard deviation of scale point variation. The p-value associated with scale point variation is 0.0000488. Thus, there is statistically significant negative correlation between Standard compliance and standard deviation of scale point variation.

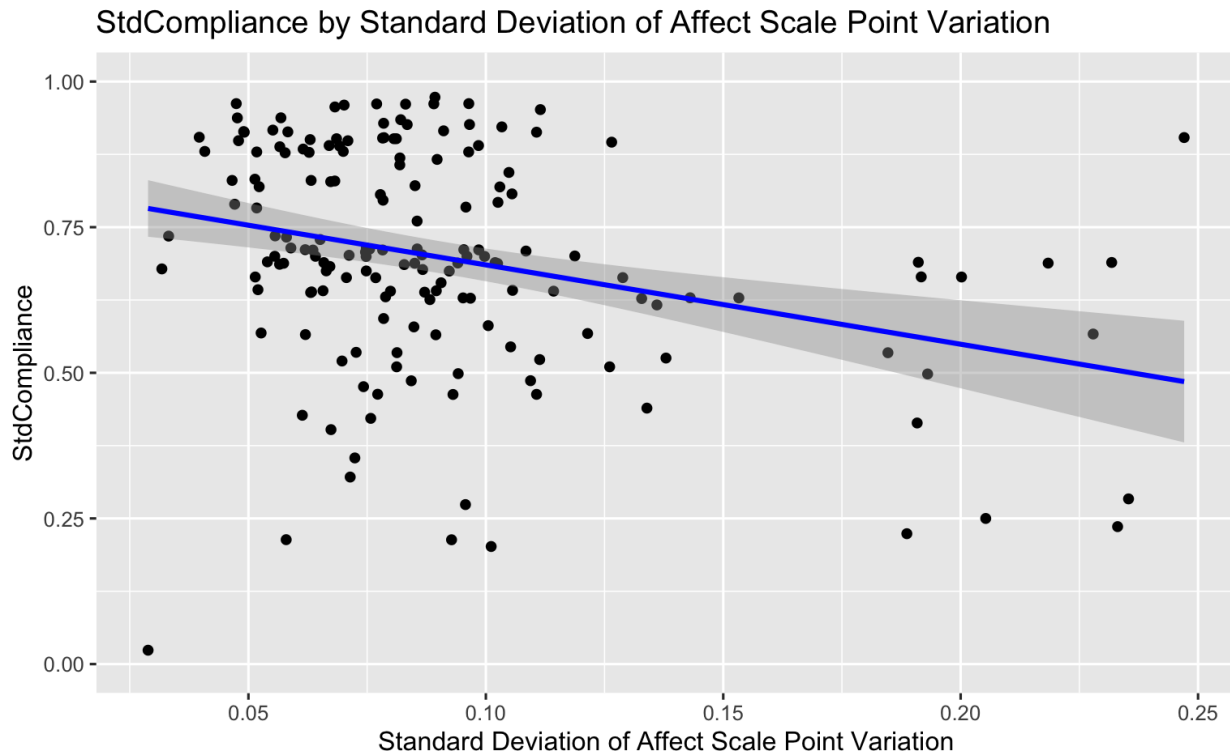


Figure 3.1.9

Overall, we see a negative correlation with compliance and standard deviation of scale point variation. This implies that variations of straightlining, measured by scale point variation, has a negative correlation with compliance. And variations of quality of EMA, indirectly measured by standard deviation of straightlining, has a negative correlation with compliance.

Moreover, the bimodal distribution of standard deviation of scale point variation shows that there are two distinct groups of individuals who have higher variation of scale point variation and lower variation of scale point variation. Hence, connecting the distribution of standard deviation of scale point variation and compliance back, we can claim that there are two distinct groups of individuals with higher variation of scale point variation and lower compliance and individuals with lower variation of scale point variation and higher compliance.

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Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233.

3.2 Survey Time and Compliance.

3.2.1 Introduction.

The Life@CMU (Phase 1) study collected survey completion times for participant EMA surveys. We investigated relationships between participants' survey completion times and their compliance during each week of the study.

Previous literature has documented that participants who have *per-question* completion times close to the median tend to have higher quality responses. Completing questions too quickly (by tapping on a random response) is called *rushing*, and rushing produces poor quality data. The same applies for participants who spend unreasonably long on a question: they were likely distracted and aren't paying full attention to the prompt. Unfortunately, the Life@CMU study was unable to collect completion times at a *question*-level; it only collected completion times at a *survey*-level.

3.2.2 Aggregation.

In our data analysis, we aggregated completion times for each week and compared it with compliance for that week. We aggregated the completion times in two different ways, resulting in **two different interpretations of aggregated completion time**:

- *Median Duration of Completed Surveys (MDCS).*
- *Median Time Spent on All Surveys (MTSS).*

Figure 3.2.1 describes how each statistic was calculated from the raw data. In the raw data, for surveys the participant didn't even attempt, an NA value was reported. In the former statistic (*Median Duration of Completed Surveys*), we remove these NA values before calculating the medians; in the latter statistic (*Median Time Spent on All Surveys*), we set these NA values to 0 before calculating the medians.

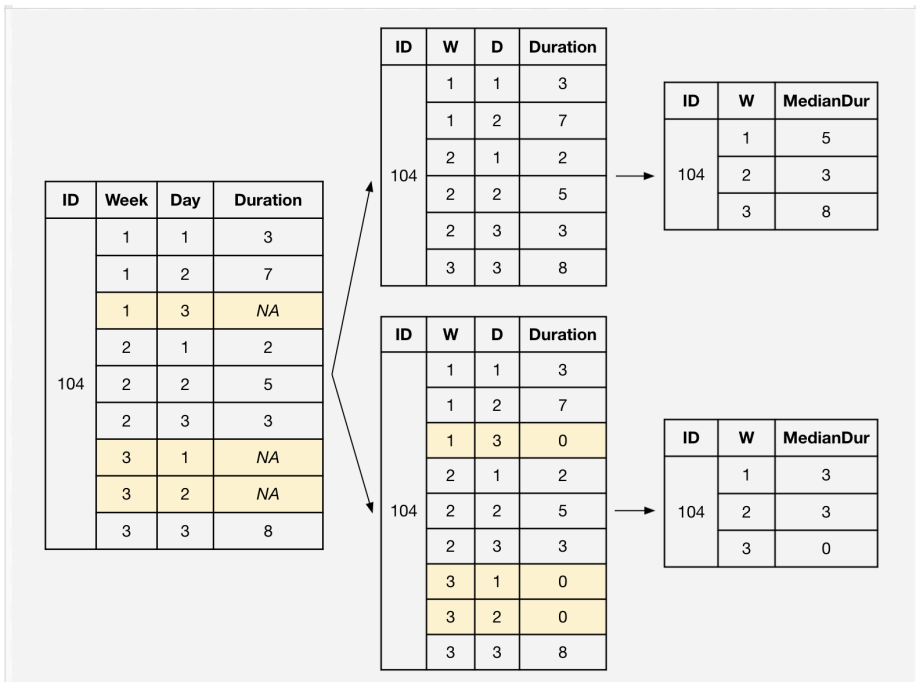


Figure 3.2.1 :: Schematic describing the 2 different aggregation techniques used on the raw survey completion time data. The left table represents the raw data, the middle table represents the two different processing techniques, and the right tables are the final aggregated data. Highlighted are the NA rows (unattempted surveys); note how they are removed in the top-middle table and set to 0 in the bottom-middle table.

3.2.3 Distributions.

We first examined the distributions of each aggregated completion time calculation. **Figure 3.2.2** and **Figure 3.2.3** describe these distributions.

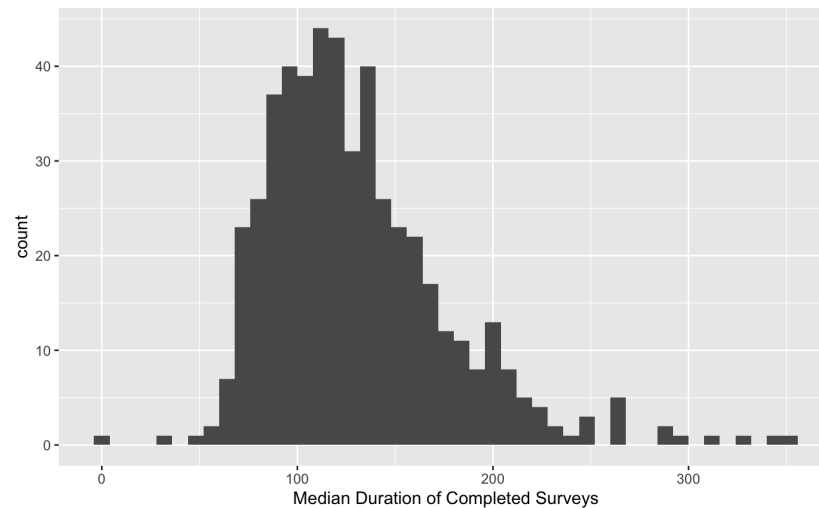


Figure 3.2.2 :: Distribution of *Median Duration of Completed Surveys* (MDCS) across all students. The distribution is shaped like a skewed Gaussian, and our count tapers off slowly as MDCS increases.

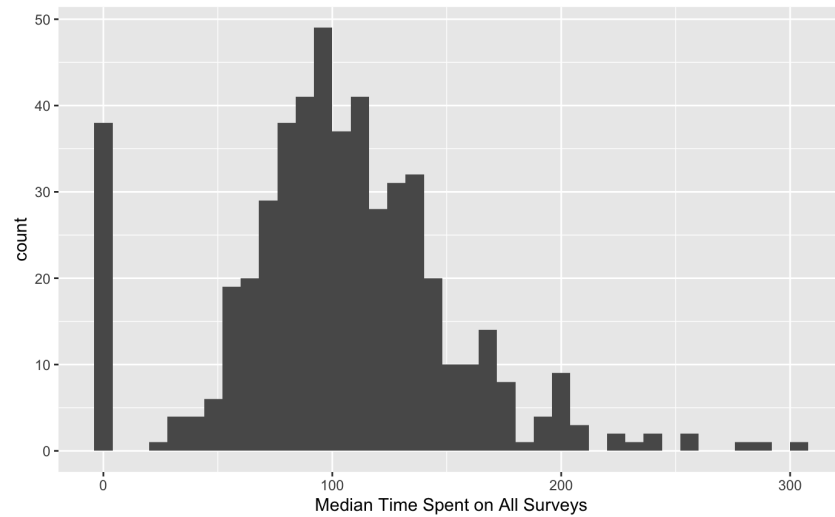


Figure 3.2.3 :: Distribution of *Median Time Spent on All Surveys (MTSS)* across all students. Excluding participants with 0 median time-spent, the distribution more closely follows Gaussian. Our bar of outliers indicates there were a significant number of students who didn't attempt most surveys during the week.

As expected and documented in the literature, these times generally follow a tight Gaussian distribution. In **Figure 3.2.3**, we see a large number of participants with 0 median time spent; combined with the absence of this bar in **Figure 3.2.2**, we can conclude a large number of participants attempted some, but very few surveys during the week.

3.2.4 Compliance Regressions.

We now investigated our aggregated survey calculations' relationship with compliance.

We first plotted StdCompliance (Y) against MDCS (X) and fit a simple least-squares regression against this data. The results are located in **Figure 3.2.4**.

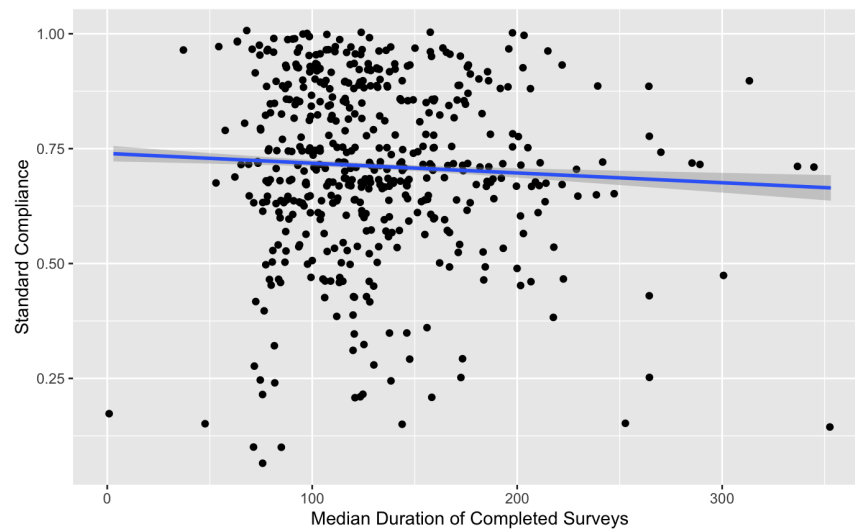


Figure 3.2.4 :: A simple linear regression (SLR) between StdCompliance (Y) and Median Duration of Completed Surveys (X). There was no statistically significant correlation between these variables under the SLR Assumptions.

This regression suggests no significant relationship between compliance and MDCS, despite a slight downward trend. Because we only take completed surveys into account, those with low compliance appear indistinguishable from those with high general compliance; thus for the surveys participants do fill out, most survey times are similar.

We now plot StdCompliance (Y) against MTSS (X) and fit a polynomial least-squares regression against this data. The results are located in **Figure 3.2.5**.

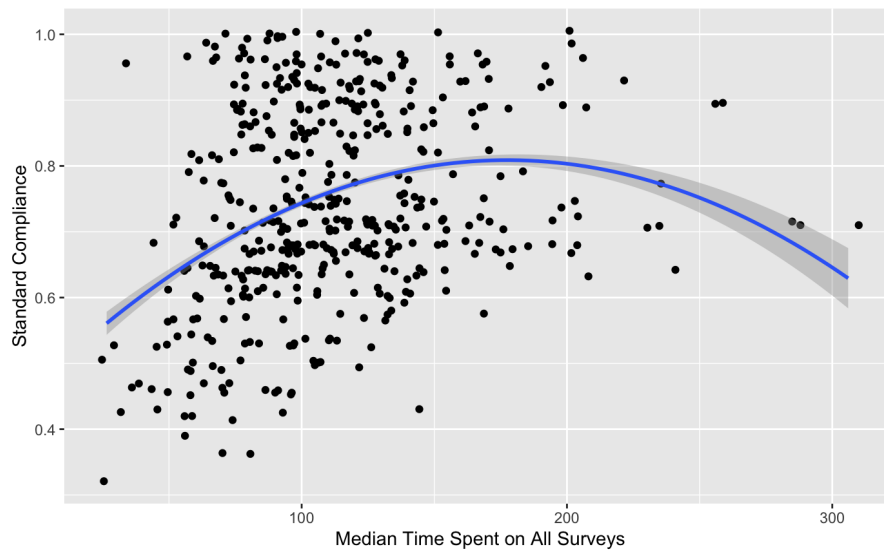


Figure 3.2.5 :: A quadratic polynomial regression between StdCompliance (Y) and Median Time Spent on All Surveys (X). There was a statistically significant correlation between the linear and quadratic factors of the model against StdCompliance.

This regression suggests a strong quadratic relationship between compliance and MDCS: those with low time-spent have low compliance, those with mean/median time spent have high compliance, and those with high time-spent have low compliance. This tends to agree more with literature findings, even though those times were collected at a finer level.

Figure 3.2.6 and **Figure 3.2.7** show the same relationship for different measures of compliance (the effect is even more pronounced).

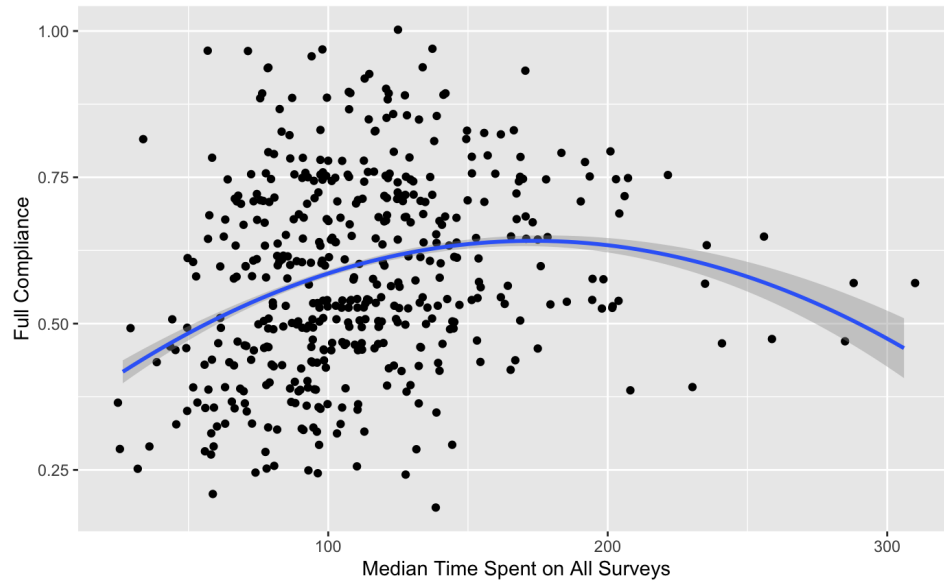


Figure 3.2.6 :: A quadratic polynomial regression between FullCompliance (Y) and Median Time Spent on All Surveys (X). There was a statistically significant correlation between the linear and quadratic factors of the model against FullCompliance.

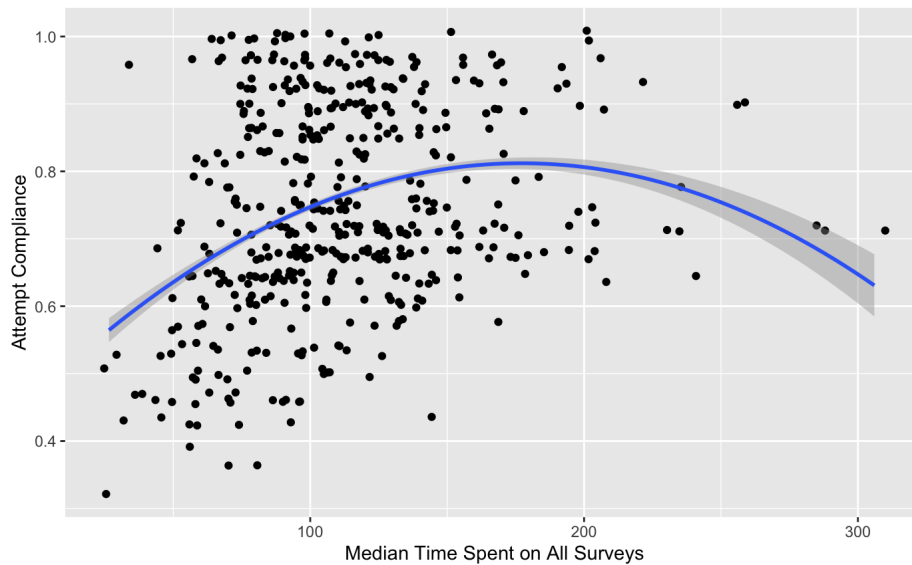


Figure 3.2.7 :: A quadratic polynomial regression between AttemptCompliance (Y) and Median Time Spent on All Surveys (X). There was a statistically significant correlation between the linear and quadratic factors of the model against AttemptCompliance.

Note that MTSS and compliance are intertwined (dependent) by design, as we are taking unattempted surveys into account (they have 0 compliance). It's likely that low compliant individuals with a mean MDCS have a much lower MTSS *due to their low compliance*. **Figure 3.2.8** shows this effect.

But the quadratic relationship found in our dataset was quite novel and will likely help us develop a time-series model.

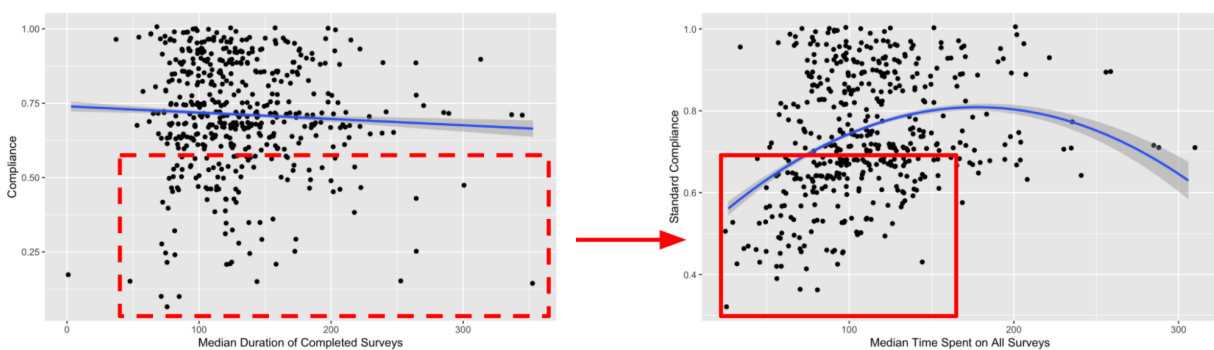


Figure 3.2.8 :: Individuals who have a higher/mean MDCS likely have a much lower MTSS because we

are accounting for uncompleted surveys. This also indicates these individuals have high response variance: they likely respond sporadically to surveys during the week. Note how high-compliance individuals are less affected by this transformation.
